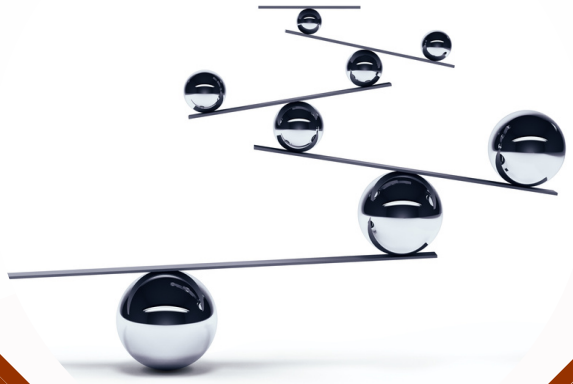
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Physical Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Ionic Equilibrium

English Medium

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IONIC EQUILIBRIUM

1. INTRODUCTION

Ionic equilibrium deals with the equilibrium of any substance with its ions in solution. The substance producing ions are called electrolytes.

According to conductivity, substances are of two types :

(i) Non-Conductor :

Those substances which do not show the flow of current or electricity.

Ex. Non - metals, plastic rubber, wood, etc.

Exception – Graphite is a non-metal but shows conductivity due to motion of free electrons.

(ii) Conductors :

Those substances which show conductivity or flow of current are called conductors. These are of 2 types :

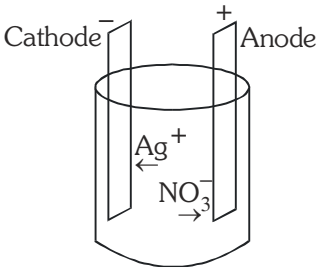
(a) Metallic or electronic conductors :

Those conductor which show conductivity due to motion of free electrons. Resistance increases with temperature.

Ex. All metals, Graphite

(b) Ionic or electrolytic conductors :

Those conductors which show conductivity due to movement of free ions. Ions are in free state in the solutions of ionic compounds. On passing electric current through the solution, ions move towards oppositely charged electrodes, i.e., the cation moves towards cathode (negative electrode) and the anion moves towards anode (positive electrode).



The current flows through the solution due to the movement of the ions. Resistance decrease with temperature.

According to strength, ionic conductors are of two types :

(i) Strong electrolyte : Those ionic conductors which are completely ionized in aqueous solution are called as strong electrolyte.

For strong electrolyte the value of degree of dissociation is 100%.

i.e. : $\alpha = 1$

Ex. (a) Strong acid $\rightarrow$ H_2SO_4, HCl, HNO_3 $HClO_4$, H_2SO_5, HBr, HI, $HBrO_4$, HIO_4, RSO_3H

(b) Strong base $\rightarrow$ KOH, NaOH, $Ba(OH)_2$ CsOH, RbOH

(c) All soluble salts $\rightarrow$ NaCl, KCl, $CuSO_4$.....

(ii) Weak electrolytes : Those electrolytes which are partially ionized in aqueous solution are called as weak electrolytes. For weak electrolytes the value of α is less than one.

Ex. (a) Weak acid : HCN, CH_3COOH, HCOOH, H_2CO_3, H_3PO_3, H_3PO_2, etc.

(b) Weak base : NH_4OH, $Cu(OH)_2$, $Zn(OH)_2$, $Fe(OH)_3$, $Al(OH)_3$, etc.

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1.2 DEGREE OF DISSOCIATION / IONISATION

- When an electrolyte is dissolved in a solvent (H_2O), it spontaneously dissociates into ions.
- It may dissociate partially ($\alpha < 1$) or sometimes completely ($\alpha \cong 1$)
- The degree of dissociation (α) of an electrolyte is the fraction of mole of the electrolyte that has dissociated under the given conditions.

$$\alpha = \frac{\text{No. of moles dissociated}}{\text{No. of moles taken initially}}$$

1.3 FACTORS AFFECTING THE VALUE OF DEGREE OF DISSOCIATION:

(i) Dilution : $\alpha \propto \sqrt{V}$. So on dilution, α increases

(ii) Temperature : On increasing temperature, ionization increases so, α increases

(iii) Nature of electrolyte :

(a) Strong electrolyte $\alpha = 100\ \%$

(b) Weak elecrolyte $\alpha << 100\ \%$

(iv) Nature of solvent :

If dielectric constant, μ , of solvent increases, then the value of α increases.

$H_2O \rightarrow \mu = 81$

$D_2O \rightarrow \mu = 79$

$C_6H_6 \rightarrow \mu = 2.5$

$CCl_4 \rightarrow \mu = 0$

Ex.1 Which one has greater α_1 or α_2 for the following equation :

(i) $NH_4OH + H_2O \rightarrow \alpha_1$

(ii) $NH_4OH + D_2O \rightarrow \alpha_2$

Sol. Dielectric constant of H_2O is more than that of D_2O , so $\alpha_1 > \alpha_2$

(v) Mixing of Ions :

Common ion Effect	Odd ion Effect
$NH_4OH \rightleftharpoons NH_4^+ + OH^-$	$NH_4OH \rightleftharpoons NH_4^+ + OH^-$
On mixing NH_4Cl	On mixing HCl
$NH_4Cl \rightarrow NH_4^+ + Cl^-$	$HCl \rightarrow H^+ + Cl^-$
Due to mixing of common ion, concentration of ammonium ion will increase therefore equilibrium will shift in backward direction means α decreases.	Due to reaction of OH^- ions with H^+ ion, concentration of OH^- will decrease $\therefore$ Equilibrium will shift in forward direction means α increases.

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1.4 OSTWALD'S DILUTION LAW (FOR WEAK ELECTROLYTE)

- For a weak electrolyte A^+B^- dissolved in water, if α is the degree of dissociation then

	$AB \rightleftharpoons$	A^+	+	B^-
initial conc.	C M	0		0
conc-at eq.	$C(1-\alpha)M$	$C\alpha M$		$C\alpha M$

Then according to law of mass action,

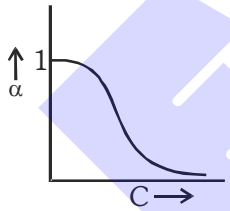
$K_{diss} = \frac{[A^+][B^-]}{[AB]} = \frac{C\alpha.C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)}$ = dissociation constant of the weak electrolyte.

$\left[C = \frac{1}{V}$, then V = 1/C (volume of solution in which 1 mole is present) is called dilution, so $K_{diss} = \frac{\alpha^2}{(1-\alpha)V} \right]$

If α is negligible in comparison to unity, $1-\alpha \simeq 1$. so $K_{diss} = \alpha^2 C \Rightarrow \alpha = \sqrt{\frac{K_{diss}}{C}} = \sqrt{K_{diss} V}$

$$\alpha \propto \frac{1}{\sqrt{\text{concentration}}}$$

- As concentration increases $\Rightarrow$ α decreases
- At infinite dilution α reaches its maximum value, unity.



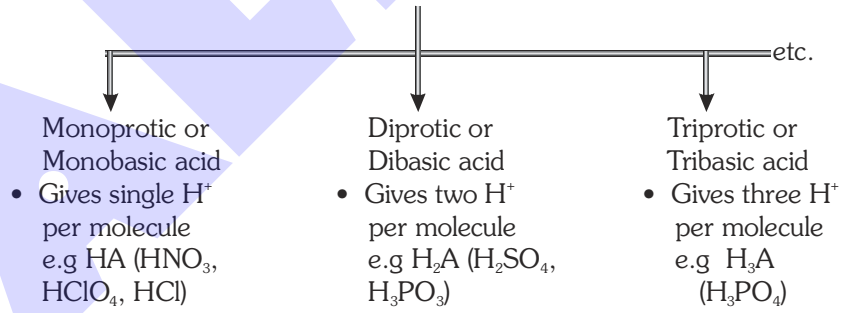
2. ACIDS BASES AND SALTS

2.1 Arrhenius concept :

(i) **Arrhenius Acid :** Substance which gives H^+ ion on dissolving in water (H^+ donor)

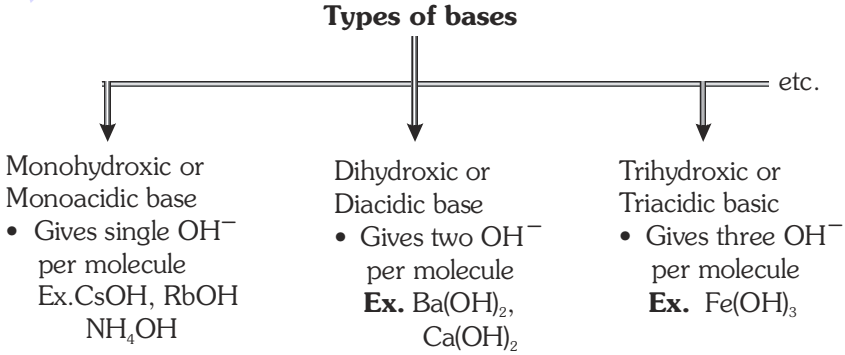
Ex. HNO_3, $HClO_4$, HCl, HI, HBr, H_2SO_4, H_3PO_4 etc.

Types of acids



(ii) Arrhenius base : Any substance which releases OH^- (hydroxyl) ion in water (OH^- ion donor)

Types of bases



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(iii) Strength of Acid or Base :

(a) Strength of acids or bases depends on the extent of its ionisation. Hence equilibrium constant K_a or K_b respectively of the following equilibria give a quantitative measurement of the strength of the acid or base.

(b) $HA \rightleftharpoons H^+ + A^-$;

$K_a = \frac{[H^+][A^-]}{[HA]}$ = dissocation or ionisation constant of acid.

(c) Similarly

$BOH \rightleftharpoons B^+ + OH^-$;

$K_b = \frac{[B^+][OH^-]}{[BOH]}$ = dissocation or ionisation constant of base

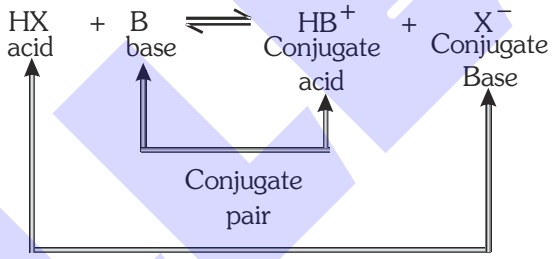
(d) Larger the value of K_a or K_b, stronger is the acid or base respectively.

2.2 Bronsted - Lowry concept : (Conjugate acid - base concept) (Protonic concept)

(i) Acid : substances which donate H^+ are Bronsted Lowry acids (H^+ donor)

(ii) Base : substances which accept H^+ are Bronsted Lowry bases (H^+ acceptor)

(iii) Conjugate acid - base pairs : In a typical acid base reaction, $HX + B \rightleftharpoons X^- + HB^+$



- Forward reaction - Here HX being a proton donor is an acid
  B being a proton acceptor is a base.
- Backward reaction - Here HB^+ being a proton donor is an acid
  X^- being a proton acceptor is a base.

	Acid		Base		Conjugate Acid		Conjugate Base
•	HCl	+	H_2O	$\rightleftharpoons$	H_3O^+	+	Cl^-
•	HSO_4^-	+	NH_3	$\rightleftharpoons$	NH_4^+	+	SO_4^{-2}
•	$[Fe(H_2O)_6]^{3+}$	+	H_2O	$\rightleftharpoons$	H_3O^+	+	$[Fe(H_2O)_5(OH)]^{2+}$

- Conjugate acid - base pair differ by only one proton.
- Strong acid will have weak conjugate base and vice versa.

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Acid	Conjugate base	Base	Conjugate acid
HCl	Cl^-	NH_3	NH_4^+
H_2SO_4	HSO_4^-	H_2O	H_3O^+
HSO_4^-	SO_4^{2-}	RNH_2	RNH_3^+
H_2O	OH^-		

(iv) Amphoteric (amphiprotic) : Substances which can act as acid as will as base are known as amphoteric

$$HCl + H_2O \rightleftharpoons H_3O^+ + Cl^-$$

base

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

acid

(v) Classification of Bronsted - Lowery Acids and Bases :

Bronsted - Lowery acids and bases can be

(i) Molecular (ii) Cationic and (iii) Anionic

Table - 1

Type	Acid	Base
Molecular	HCl, HNO_3, $HClO_4$,	NH_3, N_2H_4, Amines,
	H_2SO_4, H_3PO_4, H_2O etc.	H_2O, Alcohol, Ethers, etc.
Cationic	NH_4^+, $N_2H_5^+$, PH_4^+,	$[Fe(H_2O)_5OH]^{2+}$
	$[Fe(H_2O)_6]^{3+}$, $[Al(H_2O)_6]^{3+}$ etc.	$[Al(H_2O)_5OH]^{2+}$ etc.
Anionic	HS^-, HSO_3^-, $H_2PO_4^-$,HSO_4^-	Cl^-, Br^-, OH^-, HSO_4^-, CN^-,
	HCO_3^-, HPO_4^{2-}, etc.	CO_3^{2-},SO_4^{2-},NH_2^-, CH_3COO^- ,etc.
	all amphiprotic anions	all amphiprotic anions

2.3 Lewis concept (electronic concept) :

(i) Acid : An acid is a molecule/ion which can accept an electron pair with the formation of a coordinate bond.

Ex. Electron deficient molecules : BF_3, $AlCl_3$, etc.

Cations : H^+, Fe^{2+}, Na^+, etc.

Molecules with vacant orbitals : SF_4, PF_3

(ii) Base : A base is any molecule/ion which has a pair of electrons which can be donated.

Ex. Molecules with lone pairs : NH_3, PH_3, H_2O, CH_3OH

Anions : OH^-, H^-, NH_2^-, etc.

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3. PROPERTIES OF WATER

(i) Molar concentration / Molarity of water :

Molarity = No. of moles/litre = $\frac{1000\,g/litre}{18\,g/mole}$ = 55.55 mole/litre **= 55.55 M** (density = 1 g/cc)

(ii) Ionic product of water :

According to arrhenius concept, $H_2O \rightleftharpoons H^+ + OH^-$

So, ionic product of water, $K_w = [H^+][OH^-] = 10^{-14}$ at 25° (experimental)

Dissociation of water is endothermic, so on increasing temperature K_w increases.

(iii) Degree of dissociation of water :

$$H_2O \rightleftharpoons H^+ + OH^- \Rightarrow \alpha = \frac{\text{decrease in concentration}}{\text{initially concentration}}$$

$$= \frac{10^{-7}}{55.55} = 18 \times 10^{-10} \text{ or } 1.8 \times 10^{-7}\,\% \qquad [\text{at } 25°C]$$

(iv) Dissociation or ionisation constant of water :

$$H_2O \rightleftharpoons H^+ + OH^- \qquad K_a = K_b = \frac{[H^+][OH^-]}{[H_2O]} = \frac{10^{-7} \times 10^{-7}}{55.55} = 1.8 \times 10^{-16}$$

So, $pK_a = pK_b = -\log(1.8 \times 10^{-16}) = 16 - \log 1.8 = 15.74$

Ex.2. At dissociation constant of heavy water is 4×10^{-15} at 35°C. If its density is 1.04 g/mL. Calculate its ionic product & degree of dissociation.

Sol. $K_w = K_d[D_2O] = \left(4 \times 10^{-15} \times \frac{1040}{20}\right) = 2.08 \times 10^{-13}$

$$d = \sqrt{\frac{K_w}{C}} = \sqrt{\frac{2.08 \times 10^{-13}}{52}} = 12.64 \times 10^{-8}$$

Ex.3 Calculate ionic product of H_2O at 50°C.

Sol. $\Delta H = 13.7 \times 10^3$ cal

$$\log \frac{K_2}{10^{-14}} = \frac{13.7 \times 10^3}{2}\left(\frac{1}{298} - \frac{1}{323}\right)$$

Ex.4 The hydronium ion conc. in an aq. H_2CO_3 solution is 4×10^{-4} M at 25°C OH^- ion conc. in the solution is :

(A) 0 ***(B) 2.5×10^{-10}*** ***(C) 2.5×10^3*** ***(D) 2.5×10^{-11} M***

Answer :(D)

Ex.5 ***Select the correct option from the following ?***

(A) pK_W increases with increase of temperature

(B) pK_W decreases with increase of temperature

(C) $pK_W = 14$ at all temperatures

(D) $pK_W = pH$ at all temperatures

Sol. (B)

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4. Acidity and pH scale :

(i) Acidic strength means the tendency of an acid to give H_3O^+ or H^+ ions in water.

So greater then tendency to give H^+, more will be the acidic strength of the substance.

(ii) Basic strength means the tendency of a base to give OH^- ions in water.

So greater the tendency to give OH^- ions, more will be basic strength of the substance.

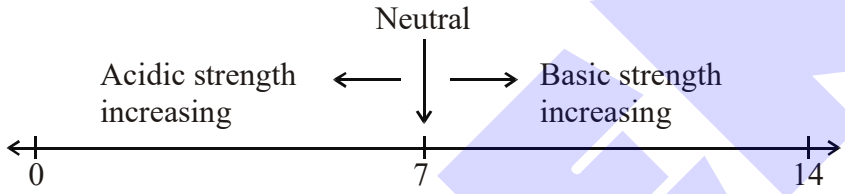
(iii) The concentration of H^+ ions is written in a simplified form introduced by **Sorenson** known as pH scale. pH is defined as negative logarithm of activity of H^+ ions.

$\therefore$ **$pH = -\log a_{H^+}$** (where a_{H^+} is the activity of H^+ ions)

(iv) Activity of H^+ ions is the molar concentration of free H^+ ions or H_3O^+ ions in a dilute solution, but unitless.

(v) Now $pH = -\log[H^+] = 7$ and $pOH = -\log[OH^-] = 7$ for water at 25°C (experimental)

$pH = 7 = pOH \Rightarrow$ neutral
$pH < 7$ or $pOH > 7 \Rightarrow$ acidic
$pH > 7$ or $pOH < 7 \Rightarrow$ Basic
} at 25 °C



4.1 pH Calculation of different Types of solutions :

(a) Strong acid solution :

(i) If concentration of H^+ ions is greater than 10^{-6} M, H^+ ions coming from water can be neglected,

So $[H^+]$ = normality of strong acid solution

(ii) If concentration is less than 10^{-6} M, H^+ ions coming from water cannot be neglected. So

$[H^+]$ = normality of strong acid + H^+ ions coming from water in presence of this strong acid

Ex.6 Calculate pH of 10^{-8} M HCl solution.

Sol. $H_2O \rightleftharpoons H^+ + OH^-$

$10^{-8}+x \quad x$

$k_w = [H^+][OH^-]$

$10^{-14} = x(x + 10^{-8})$

$\Rightarrow \quad x^2 + x \times 10^{-8} - 10^{-14} = 0$

$$x = \frac{-10^{-8} \pm \sqrt{10^{-16} + 4 \times 10^{-14}}}{2} = \frac{-10^{-8} + 10^{-7}\sqrt{4 + \frac{1}{100}}}{2} = \frac{(\sqrt{401} - 1)10^{-8}}{2} = 0.95 \times 10^{-7}$$

$[H^+] = 10.5 \times 10^{-8} = 1.05 \times 10^{-7}$

$pH = -\log [H^+]$

$pH = 7 - \log 1.05 \approx 6.98$

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4.2 Strong base solution :

Calculate the $[OH^-]$ which will be equal to normality of the strong base solution and then use

$K_w = [H^+] \times [OH^-] = 10^{-14}$, to calculate $[H^+]$

Ex.7 Calculate pH of 10^{-7} M of NaOH solution

Sol. $[OH^-]$ from NaOH = 10^{-7}

$[OH^-]$ from water = $x < 10^{-7}$ M (due to common ion effect)

$$H_2O \rightleftharpoons OH^- + H^+$$
$$- \quad (x + 10^{-7}) \quad x$$

$K_w = [H^+]\,[OH^-] = 10^{-14} = x\,(x + 10^{-7})$

$x^2 + 10^{-7}x - 10^{-14} = 0$

$$\Rightarrow \quad x = \frac{\sqrt{5}-1}{2} \times 10^{-7} = 0.618 \times 10^{-7} \qquad (\sqrt{5} = 2.236)$$

$[OH^-] = 10^{-7} + 0.618 \times 10^{-7} = 1.618 \times 10^{-7}$

$pOH = 7 - \log(1.618) = 6.79$

$pH = 14 - 6.79 = 7.21$

4.3 pH of mixture of two strong acids : If V_1 volume of a strong acid solution of normality N_1 is mixed with V_2 volume of another strong acid solution of normality N_2, then

Number of H^+ ions from I-solution = N_1V_1

Number of H^+ ions from II-solution = N_2V_2

If final normality is N and final volume is V, then

$NV = N_1V_1 + N_2V_2$

[dissociation equilibrium of none of these acids will be disturbed as both are strong acid]

$$[H^+] = N = \frac{N_1V_1 + N_2V_2}{V_1 + V_2}$$

[where $N = M \times n$, n = Basicity of acid]

4.4 pH of mixture of two strong bases :

Similar to above calculation

$$[OH^-] = N = \frac{N_1V_1 + N_2V_2}{V_1 + V_2} \qquad [H^+] = \frac{10^{-14}}{[OH^-]}$$

[where $N = M \times n$, n = Acidity of base]

Ex.8 Calculate pH of mixture of (400 mL, $\frac{1}{200}$ M H_2SO_4) + (400 mL, $\frac{1}{100}$ M HCl) + (200 mL of water)

Sol. $N_1V_1 = \frac{1}{200} \times \frac{400}{1000} \times 2 = \frac{4}{1000}$, $N_2V_2 = \frac{4}{1000}$, H^+ ions from water will be neglected

$N_1V_1 + N_2V_2 = 8 \times 10^{-3}$ $\qquad [H^+] = \frac{8 \times 10^{-3}}{1} = 8 \times 10^{-3}$

$pH = 3 - \log 8 = 2.1$

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Ex.9 500 mL of 10^{-5} M NaOH is mixed with 500 mL of 2.5×10^{-5} M of $Ba(OH)_2$. To the resulting solution 99 L water is added. Calculate pH.

Sol. $[OH^-] = \frac{500 \times 10^{-5} + 500 \times 2 \times 2.5 \times 10^{-5}}{1000} = 3 \times 10^{-5}$ M

$M_1 = 3 \times 10^{-5}$ M

$V_2 + V_1 = 1$ L

$V_F = 100$ L

no. of moles of $[OH^-]$ initially = no. of moles of $[OH^-]$

$3 \times 10^{-5} = M_2 \times 100$

$\because$ $M_2 = 3 \times 10^{-7} < 10^{-6}$

$H_2O \rightleftharpoons H^+ + OH^-$

$x \quad (x + 3 \times 10^{-7})$

$K_w = x\,(x + 3 \times 10^{-7}) = 10^{-14}$

$\because$ $x = \left(\frac{\sqrt{13} - 3}{2}\right) \times 10^{-7}$

$x = 0.302 \times 10^{-7}$

$[OH^-]_{Net} = 3.302 \times 10^{-7}$

4.5 pH of mixture of a strong acid and a strong base :

- Acid Base neutralisation reaction will take place.
- The solution will be acidic or basic depending on which component has been taken in excess.
- If V_1 volume of a strong acid solution of normality N_1 is mixed with V_2 volume of a strong base solution of normality N_2, then

  Number of H^+ ions from I-solution = N_1V_1

  Number of OH^- ions from II-solution = N_2V_2

If $N_1V_1 > N_2V_2$

$[H^+] = N = \frac{N_1V_1 - N_2V_2}{V_1 + V_2}$

Solution will be acidic in nature

If $N_2V_2 > N_1V_1$

$[OH^-] = N = \frac{N_2V_2 - N_1V_1}{V_1 + V_2}$

Solution will be basic in nature

$[H^+] = \frac{10^{-14}}{[OH^-]}$

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Ex.10 Calculate pH of mixture of (400 mL, $\frac{1}{200}$ M $Ba(OH)_2$) + (400 mL, $\frac{1}{50}$ M HCl)+(200 mL of water)

Sol. $[H^+] = \frac{400\times\frac{1}{50} - 400\times\frac{1}{200}\times 2}{1000} = 4\times10^{-3}$, so pH = 3 – 2 log 2 = 2.4

Ex.11 What will be the resultant pH when 150 mL of an aqueous solution of HCl (pH = 2.0) is mixed with 350 mL of an aqueous solution of NaOH (pH = 12.0) ?

Sol. pH of HCl = 2

$\therefore$ $[HCl] = 10^{-2}$ M

pH of NaOH = 12, pOH = 2 $\therefore$ $[NaOH] = 10^{-2}$ M

	HCl	+ NaOH	$\longrightarrow$ NaCl	+ H_2O
Meq. initial	150×10^{-2} = 1.5	350×10^{-2} = 3.5	0	0
Meq. final	0	2	1.5	1.5

$\therefore$ $[OH^-]$ from NaOH = $\frac{2}{500} = 4\times10^{-3}$ M

$pOH = -\log[OH^-] = -\log(4\times10^{-3})$

$\therefore$ pOH = 2.3979

$\therefore$ pH = 14 – pOH = 14 – 2.3979 = 11.6021

4.6 pH of a weak acid or weak base (monoprotic) Solution :

- Weak acid does not dissociated 100 % therefore we have to calculate the percentage dissociation using K_a dissociation constant of the acid.
- We have to use Ostwald's Dilution law (as have been derived earlier)

	HA $\rightleftharpoons$	H^+ +	A^-	
t = 0	C	0	0	
t_{eq}	C(1 – α)	Cα	Cα	$K_a = \frac{[H^+][A^-]}{[HA]} = \frac{C\alpha^2}{1-\alpha}$

If $\alpha << 1 \Rightarrow (1-\alpha)\approx 1 \Rightarrow K_a \approx C\alpha^2 \Rightarrow \alpha = \sqrt{\frac{K_a}{C}}$ (is valid if $\alpha < 0.1$ or 10%)

$[H^+] = C\alpha = C\sqrt{\frac{K_a}{C}} = \sqrt{K_a\times C}$ So $pH = \frac{1}{2}(pK_a - \log C)$

On increasing the dilution $\Rightarrow$ C↓ = α↑ and $[H^+]\downarrow \Rightarrow$ pH↑

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Ex.12 ***Calculate pH of : (a) 10^{-1} M CH_3COOH (b) 10^{-3} M CH_3COOH (c) 10^{-6} M CH_3COOH***
Take $K_a = 2 \times 10^{-5}$

Sol. **(a)** $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

C	0	0
$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$K_a = \frac{C\alpha^2}{1-\alpha} \Rightarrow \alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{2\times10^{-5}}{10^{-1}}} = \sqrt{2\times10^{-4}} \quad (\alpha << 0.1)$$

$$\text{So, } [H^+] = 10^{-1} \times \sqrt{2} \times 10^{-2} \Rightarrow pH = 3 - \frac{1}{2}\log 2 = 2.85$$

(b) $\alpha = \sqrt{\frac{K_a}{C}} \Rightarrow \alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{2\times10^{-5}}{10^{-3}}} = \sqrt{2\times10^{-2}} \quad (\alpha > 0.1)$

So we have to do the exact calculations

$$K_a = \frac{C\alpha^2}{1-\alpha} \Rightarrow 2\times10^{-5} = \frac{10^{-3}\times\alpha^2}{1-\alpha} \Rightarrow \alpha = 13.14\ \%$$

$$[H^+] = 10^{-3}\times0.1314 = 1.314\times10^{-4} \Rightarrow pH = 4 - \log(1.314) \approx 3.8$$

(c) If approximation is used the, $\alpha = \sqrt{\frac{2\times10^{-5}}{10^{-6}}} = \sqrt{20} > 1$,

So we have to do the exact calculation, $2\times10^{-5} = 10^{-6}\frac{\alpha^2}{1-\alpha} \Rightarrow \alpha \approx 0.95$ or 95%

$$[H^+] = 0.95\times10^{-6} = 9.5\times10^{-7} \Rightarrow pH = 7 - \log(9.5) = 6.022$$

- At very low concentration (at infinite dilution) weak electrolyte will be almost 100% dissociate, so behave as strong electrolyte.
(pH) of 10^{-6} M HCl ≃ pH of 10^{-6} M CH_3COOH ≃ 6)

Ex.13 K_a for acid HA is 2.5×10^{-8} calculate for its decimolar solution at 25°C.
(i) % dissociation (ii) pH (iii) OH^- ion concentration

Sol. $HA \rightleftharpoons H^+ + A^-$

C	0	0
$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow K_a = \frac{C\alpha.C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)} \approx C\alpha^2$$

(i) $\therefore \quad \alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{2.5\times10^{-8}}{1/10}}$ (C = 1/10 M)

$= 5\times10^{-4} = 0.05\%$

(ii) $[H^+] = C\alpha = \frac{1}{10}\times5\times10^{-4} = 5\times10^{-5}$ mol/L

So pH = 5 – log 5 = 4.30

(iii) $[H^+]\,[OH^-] = 1\times10^{-14}$

$\therefore \quad [OH^-] = \frac{10^{-14}}{5\times10^{-5}} = 2\times10^{-10}$ mol/L

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Ex.14 ***Determine the degree of dissociation of 0.05 M NH_4OH at 25°C in a solution of pH = 10.***

Sol. $NH_4OH \rightleftharpoons NH_4^+ + OH^-$

C 0 0

Given, pH = 10

$[H^+] = 10^{-10}$

$[H^+]\,[OH^-] = 1 \times 10^{-14}$

$\therefore \quad [OH^-] = \frac{1\times10^{-14}}{10^{-10}} = 10^{-4} = C\alpha$

$\therefore \quad \alpha = \frac{[OH^-]}{C} = \frac{10^{-4}}{0.05} = 2 \times 10^{-3}$ or 0.2 %

Ex.15 ***The concentration of $[H^+]$ and $[OH^-]$ of the 10^{-1} M aqueous solution of 2% ionised weak acid is :***

(A) 2×10^{-3} M and 5×10^{-12} M ***(B) 1×10^{-3} M and 3×10^{-11} M***

(C) 2×10^{-4} M and 5×10^{-11} M ***(D) 3×10^{-2} M and 4×10^{-13} M***

Sol. **(A)**

$[H^+] = C\alpha = 2 \times 10^{-3}$ M or $[OH^-] = \frac{10^{-14}}{[H^+]} = 5 \times 10^{-12}$ M

Ex.16 ***When a 0.1 N solution of an acid at 25°C has a degree of ionisation of 4%, the concentration of OH^- present is :***

(A) 2.5×10^{-3} ***(B) 2.5×10^{-11}*** ***(C) 2.5×10^{-12}*** ***(D) 2.5×10^{-13}***

Sol. **(C)**

$[H^+] = C\alpha = 0.1 \times 4 \times 10^{-2} = 4 \times 10^{-3}$ M or $[OH^-] = \frac{10^{-14}}{[H^+]} = 2.5 \times 10^{-12}$ N

Ex.17 ***The degree of dissociation of acetic acid in a 0.1 M solution is 1.32×10^{-2}. Calculate dissociation constant of acid and its pK_a value :***

Sol.

	CH_3COOH	$\rightleftharpoons$	CH_3COO^-	+	H^+
Initially	0.1		0		0
at equilibrium	0.1(1–0.0132)		0.1× 0.0132		0.1× 0.0132

$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{0.1\times0.0132\times0.1\times0.0132}{0.1(1-0.0132)} = 1.76 \times 10^{-5}$

$pK_a = -\log K_a = -\log (1.76 \times 10^{-5}) = 4.75$

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4.7 pH of a mixture of weak acid (monoprotic) and a strong acid solution :

- Weak acid and Strong acid both will contribute H^+ ion.
- For the first approximation we can neglect the H^+ ions coming from the weak acid solution and calculate the pH of the solution from the concentration of the strong acid only.
- To calculate exact pH, we have to take the effect of presence of strong acid on the dissociation equilibrium of the weak acid.
- If $[SA] = C_1$ and $[WA] = C_2$, then $[H^+]$ from $SA = C_1$
  the weak acid will dissociate as follows.

$$\begin{array}{ccccc} HA & \rightleftharpoons & H^+ & + & A^- \\ C_2 & & 0 & & 0 \\ C_2(1-\alpha) & & C_2\alpha + C_1 & & C_2\alpha \end{array} \qquad K_a = \frac{(C_2\alpha + C_1)C_2\alpha}{C_2(1-\alpha)} \quad (\alpha <<< 1)$$

(The weak acids dissociation will be further suppressed because of presence of strong acid, common ion effect)

$K_a = (C_2\alpha + C_1)\alpha$

Total H^+ ion concentration $= C_1 + C_2\alpha$

- If the total $[H^+]$ from the acid is more than 10^{-6}M, then contribution from the water can be neglected, if not then we have to take $[H^+]$ from the water also.

4.8 pH of a mixture of two weak acid (both monoprotic) solution :

- Both acids will dissociate partially.
- Let the acid are HA_1 & HA_2 and their final concentrations are C_1 & C_2 respectively, then

$$\begin{array}{cccccc} & HA_1 & \rightleftharpoons & H^+ & + & A_1^- \\ t=0 & C_1 & & 0 & & 0 \\ \text{At eq.} & C_1(1-\alpha_1) & & C_1\alpha_1 + C_2\alpha_2 & & C_1\alpha_1 \end{array} \qquad K_{a_1} = \frac{C_1\alpha_1(C_1\alpha_1 + C_2\alpha_2)}{C_1(1-\alpha_1)}$$

$$\begin{array}{cccccc} & HA_2 & \rightleftharpoons & H^+ & + & A_2^- \\ t=0 & C_2 & & 0 & & 0 \\ \text{At eq.} & C_2(1-\alpha_2) & & C_2\alpha_2 + C_1\alpha_1 & & C_2\alpha_2 \end{array} \qquad K_{a_2} = \frac{(C_2\alpha_2 + C_1\alpha_1)C_2\alpha_2}{C_2(1-\alpha_2)}$$

(Since α_1, α_2 both are small in comparison to unity)

$$K_{a_1} = (C_1\alpha_1 + C_2\alpha_2)\alpha_1 \; ; \; K_{a_2} = (C_1\alpha_1 + C_2\alpha_2)\alpha_2 \quad \Rightarrow \quad \frac{K_{a_1}}{K_{a_2}} = \frac{\alpha_1}{\alpha_2}$$

$$[H^+] = C_1\alpha_1 + C_2\alpha_2 = \frac{C_1K_{a_1}}{\sqrt{C_1K_{a_1} + C_2K_{a_2}}} + \frac{C_2K_{a_2}}{\sqrt{C_1K_{a_1} + C_2K_{a_2}}} \Rightarrow \quad [H^+]$$

$$= \sqrt{C_1K_{a_1} + C_2K_{a_2}}$$

- If the dissociation constant of one of the acid is very much greater than that of the second acid then contribution from the second acid can be neglected.

So, $[H^+] = C_1\alpha_1 + C_2\alpha_2 \approx C_1\alpha_1$

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Ex.18 ***Calculate pH of solution obtained by mixing equal vol. of 0.02 M HOCl & 0.2 M CH_3COOH solution given that K_{a_1} (HOCl) = 2×10^{-4} , K_{a_2} (CH_3COOH) = 2×10^{-5}***

Also calculate OH^-, OCl^-, CH_3COO^-

Sol. Final solution volume become double

$C_1 = 0.01, \quad C_2 = 0.1$

$[H^+] = \sqrt{K_{a_1}C_1 + K_{a_2}C_2} = \sqrt{2\times10^{-4}\times0.01 + 2\times10^{-5}\times0.1}$

$= \sqrt{2\times10^{-6} + 2\times10^{-6}} = 2 \times 10^{-3}$

$pH = 3 - \log 2 = 3 - 0.3010 = 2.69$

$\alpha_1 = \frac{2\times10^{-4}}{2\times10^{-3}} = 10^{-1} \quad \alpha_2 = \frac{2\times10^{-5}}{2\times10^{-3}} = 10^{-2}$

$HOCl \rightleftharpoons H^+ + OCl^-$	$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$
$C_1(1-\alpha_1) \quad C_1\alpha_1 + C_2\alpha_2 \quad C_1\alpha_1$	$C_2(1-\alpha_2) \quad C_1\alpha_1 + C_2\alpha_2 \quad C_2\alpha_2$
$[OCl^-] = C_1\alpha_1$	$[CH_3COO^-] = C_2\alpha_2$
$= 0.01 \times 10^{-1}$	$= 0.1 \times 10^{-2}$
$= 1 \times 10^{-3}$	$= 1 \times 10^{-3}$

$[OH^-] = \frac{K_w}{[H^+]} = \frac{10^{-14}}{2\times10^{-3}} = 0.5 \times 10^{-11} = 5 \times 10^{-12}$ M

$[HOCl] = 10^{-2}(1 - 0.1) = 9 \times 10^{-3}$ M

$[CH_3COOH] = 10^{-1}(1 - 0.01) \approx 10^{-1}$

4.9 pH of a solution of a polyprotic weak acid :

- Diprotic acid is the one, which is capable of giving 2 protons per molecule in water. Let us take a weak diprotic acid (H_2A) in water whose concentration is c M.

  In an aqueous solution, following equilbria exist.

If

α_1 = degree of ionization of H_2A in presence of HA^- $\quad$ K_{a_1} = first ionisation constant of H_2A

α_2 = degree of ionisation of HA^- in presence of H_2A $\quad$ K_{a_2} = second ionisation constant of H_2A

I step $\quad$ **II step**

$H_2A + H_2O \rightleftharpoons HA^- + H_3O^+$ $\quad$ $HA^- + H_2O \rightleftharpoons A^{2-} + H_3O^+$

at eq. $c(1-\alpha_1) \quad c\alpha_1(1-\alpha_2) \quad (c\alpha_1 + c\alpha_1\alpha_2)$ $\quad$ at eq. $c\alpha_1(1-\alpha_2) \quad c\alpha_1\alpha_2 \quad (c\alpha_1 + c\alpha_1\alpha_2)$

$(K_{eq})_1 [H_2O] = \frac{[H_3O^+][HA^-]}{[H_2A]} = \mathbf{K_{a_1}}$ $\quad$ $(K_{eq})_2 [H_2O] = \frac{[H_3O^+][A^{2-}]}{[HA^-]} = \mathbf{K_{a_2}}$

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$\therefore \quad K_{a_1} = \dfrac{(c\alpha_1 + c\alpha_1\alpha_2)[c\alpha_1(1-\alpha_2)]}{c(1-\alpha_1)}$ $\qquad$ $K_{a_2} = \dfrac{(c\alpha_1 + c\alpha_1\alpha_2)[c\alpha_1\alpha_2)]}{c\alpha_1(1-\alpha_2)}$

$= \dfrac{[c\alpha_1(1+\alpha_2)][\alpha_1(1-\alpha_2)]}{1-\alpha_1}$ ..... (i) $\qquad$ $= \dfrac{[c\alpha_1(1+\alpha_2)]\alpha_2}{1-\alpha_2}$ ..... (ii)

Knowing the values of K_{a_1}, K_{a_2} and c, the values of α_1 and α_2 can be calculated using equations (i) and (ii) After getting the values of α_1 and α_2, $[H_3O^+]$ can be calculated as

$\mathbf{[H_3O^+]_T = c\alpha_1 + c\alpha_1\alpha_2}$

Finally, for calculation of pH

- If the total $[H_3O^+] < 10^{-6}$ M, the contribution of H_3O^+ from water should be added.
- If the total $[H_3O^+] > 10^{-6}$ M, then $[H_3O^+]$ contribution from water can be ignored.

Using this $[H_3O^+]$, pH of the solution can be calculated.

❖ **Approximation :**

For diprotic acids, $K_{a_2} << K_{a_1}$ and α_2 would be even smaller than α_1

$\therefore \quad 1 - \alpha_2 \approx 1$ and $1 + \alpha_2 \approx 1$

Thus, equation (i) can be reduced to $K_{a_1} = \dfrac{C\alpha_1 \times \alpha_1}{1-\alpha_1}$

This is expression similar to the expression for a weak monoprotic acid.

- Hence, for a diprotic acid (or a polyprotic acid) the $[H_3O^+]$ can be calculated from its first equilibrium constant expression alone provided $K_{a_2} << K_{a_1}$.

Ex.19 Calculate pH of $[HS^-]$, $[S^{2-}]$, $[Cl^-]$ in a solution which is 0.1 M HCl & 0.1 M H_2S given that $K_{a_1}(H_2S) = 10^{-7}$, $Ka_2(H_2S) = 10^{-14}$ also calculate α_1 & α_2.

Sol. HCl + H_2S

0.1 0.1

$C_1 = C_2 = 0.1$

$\therefore \quad pH = 1$ (most of $[H^+]$ comes from HCl]

$H_2S \rightleftharpoons H^+ + HS^-$

$0.1(1-\alpha_1) \quad 10^{-1} \quad C\alpha_1 = 0.1\ \alpha_1$

$Ka_1 = \dfrac{C\alpha_1 \times 10^{-1}}{C(1-\alpha_1)} = \dfrac{10^{-7}}{10^{-1}} = \alpha_1 \ (\because 1 - \alpha_1 = 1)$

$\Rightarrow \quad \alpha_1 = 10^{-6}$

$HS^- \rightleftharpoons S^{2-} + H^+$

$C\alpha_1(1-\alpha_2) \quad C\alpha_1\alpha_2 \quad 0.1$

$10^{-14} = 0.1 \times \alpha_2$

$\Rightarrow \quad \alpha_2 = 10^{-13}$

$[S^{2-}] = C\alpha_1\alpha_2$

$= 10^{-6} \times 10^{-1} \times 10^{-13} = 10^{-20}$ M

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4.10 ISOHYDRIC SOLUTIONS

(i) Solutions of electrolytes are said to be isohydric if the concentration of the common ion present in them is the same and on mixing such solutions, there occurs no change in the degree of dissociation of either of the electrolyte.

(ii) Let the isohydric solution is made by HA_1 and HA_2 acids, then $[H^+]$ of both acids should be equal i.e.

$$\sqrt{K_{a_1} C_1} = \sqrt{K_{a_2} C_2} \qquad \text{or} \qquad \frac{K_{a_1}}{K_{a_2}} = \frac{C_2}{C_1}$$

4.11 RELATIVE STRENGTH OF WEAK ACIDS AND BASES

For two acids of equimolar concentrations.

$$\frac{\text{Strength of acid (I)}}{\text{Strength of acid (II)}} = \sqrt{\frac{K_{a_1}}{K_{a_2}}}$$

Similarly for bases, $\frac{\text{Strength of base (I)}}{\text{Strength of base (II)}} = \sqrt{\frac{K_{b_1}}{K_{b_2}}}$

The modern method is to convert K_a as a power of 10 and express acid strength by power of 10 with sign changed and call this new unit pK_a. Thus, if K_a for acid is equal to 10^{-4}, $pK_a = 4$. So higher pK_a value means lower acid strength, that is, $pK_a = -\log K_a$

Also, $pK_b = -\log K_b$

5. SALTS

(i) Salts are the ionic compounds formed when its positive part (Cation) come from a base and its negative part (Anion) come from an acid.

(ii) Salts may taste salty, bitter or sweet or tasteless.

(iii) Solution of salts may be acidic, basic or neutral.

(iv) Fused salts and their aqueous solutions conduct electricity and undergo electrolysis.

(v) The salts are generally crystalline solids.

5.1 Classification of salts :

The salts may be classified into four categories.

(a) Normal salt :

(i) The salt formed by the loss of all possible protons (replaceable H^+ ions)

Ex. NaCl, $NaNO_3$, K_2SO_4, $Ca_3(PO_4)_2$, Na_3BO_3, Na_2HPO_3, NaH_2PO_2 etc.

(b) Acid salts :

(i) Salts formed by incomplete neutralisation of polybasic acids. Such salts contain one or more replaceable H atom.

Ex. $NaHCO_3$, $NaHSO_4$, NaH_2PO_4, Na_2HPO_4 etc.

(ii) Above salts when neutralized by base form normal salts.

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(c) Basic salts :

(i) Salts formed by in complete neutralisation of poly acidic bases are called basic salts. These salt contain one or more hydroxyl groups.

Ex. $Zn(OH)Cl$, $Mg(OH)Cl$, $Fe(OH)_2Cl$, $Bi(OH)_2Cl$ etc.

(ii) Above salts when neutralised by acids form normal salts.

5.2 HYDROLYSIS OF SALTS

Salt hydrolysis is defined as the process in which water reacts with cation or anion or both of a salt to change the concentration of H^+ and OH^- ions of water.

Salt hydrolysis is reverse process of neutralization.

$$\text{Water + Salt} \rightleftharpoons \text{Acid + Base ;} \qquad \Delta H = +ve$$

5.2.1 Hydrolysis of strong acid - weak base [SA - WB] type salt –

Ex. $CaSO_4$, NH_4Cl, $(NH_4)_2SO_4$, $Ca(NO_3)_2$, $ZnCl_2$, $CuCl_2$, $CaCl_2$

$$NH_4Cl + H_2O \rightleftharpoons NH_4OH + HCl$$

$$NH_4^+ + Cl^- + H_2O \rightleftharpoons NH_4OH + H^+ + Cl^-$$

Net reaction : $NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

(i) In this type of salt hydrolysis, cation reacts with H_2O, therefore called as ***cationic hydrolysis.***

(ii) Solution is acidic in nature (SAWB) as $[H^+]$ is increased.

(iii) pH of the solution is less than 7.

(iv) **Relation between K_h , K_w & K_b**

$$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$$

Hydrolysis constant $K_h = \dfrac{[NH_4OH][H^+]}{[NH_4^+]}$ .....(i)

For weak Base $NH_4OH \rightleftharpoons NH_4^+ + OH^-$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]} \qquad \text{....(ii)}$$

For water $H_2O \rightleftharpoons H^+ + OH^-$

$K_w = [OH^-]\,[H^+]$ .....(iii)

Now multiplying Eq. (1) & (2) = Eq. (3)

$$\frac{[NH_4OH][H^+]}{[NH_4^+]} \times \frac{[NH_4^+][OH^-]}{[NH_4OH]} = [H^+]\,[OH^-]$$

i.e. $K_h \times K_b = K_w$

$$\boxed{K_h = \frac{K_w}{K_b}}$$

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(v) Degree of hydrolysis – (Represented by h)

$$\underset{C}{NH_4^+} + H_2O \rightleftharpoons \underset{0}{NH_4OH} + \underset{0}{H^+}$$

$C - Ch \qquad Ch \qquad Ch$ (initial concentration at equilibrium)

$$K_h = \frac{[NH_4OH][H^+]}{[NH_4^+]} = \frac{Ch^2}{(1-h)}$$

Since $h <<<< 1$

then $(1-h) \approx 1$

$\therefore \quad K_h = Ch^2$

$$\Rightarrow \quad h = \sqrt{\frac{K_h}{C}}$$

$$\therefore \Rightarrow \quad h = \sqrt{\frac{\frac{K_w}{K_b}}{C}} \quad \Rightarrow \quad \boxed{h = \sqrt{\frac{K_w}{K_b \times C}}}$$

(vi) pH of the solution :

$$pH = -\log [H^+]$$

$$\Rightarrow \quad [H^+] = \sqrt{\frac{K_w \times C}{K_b}}$$

On taking – log on both sides

$$pH = -\log\left(\frac{K_w \times C}{K_b}\right)^{1/2}$$

$$pH = -\frac{1}{2}\log K_w - \frac{1}{2}\log C - \frac{1}{2}(-\log K_b)$$

$$pH = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C$$

Ex.20 Find out the K_h of centi normal [10^{-2} N] solution of NH_4Cl (SA - WB) if dissociation constant of NH_4OH is 10^{-6} and $K_w = 10^{-14}$. Find out degree of hydrolysis and also find $[H^+]$ and pH of solution?

(Given : $K_w = 10^{-14}$; $K_b = 10^{-6}$)

Sol. (1) $K_h = \frac{K_w}{K_b} = \frac{10^{-14}}{10^{-6}} = 10^{-8}$

(2) $h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{10^{-8}}{10^{-2}}} = \sqrt{10^{-6}} = 10^{-3}$

(3) $[H^+] = Ch$

$= 10^{-2} \times 10^{-3} = 10^{-5}$

(4) $pH = -\log [H^+] = -\log [10^{-5}] = +5\log 10 = +5 \times 1 = 5$

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Ex.21 How many grams of NH_4Cl should be dissolved per litre of solution to have a pH of 5.13 ? K_b for NH_3 is 1.8×10^{-5}.

Sol. NH_4Cl is a salt of strong acid and weak base for solutions of such salts.

$$pH = \frac{1}{2}\,[pK_W - \log C - pK_b]$$

$\Rightarrow$ $10.26 = 14 - \log C - 4.74$

$\Rightarrow$ $\log C = 9.26 - 10.26 = -1.0$

$\therefore$ $C = 10^{-1}$ M

$[NH_4Cl] = 10^{-1}$ M

$W_{NH_4NO_3} = 10^{-1} \times 53.5\ gL^{-1}$

$= 5.35\ gL^{-1}$

5.2.2 Hydrolysis of [WA – SB] type salt :

Ex. KCN, NaCN, K_2CO_3, $BaCO_3$, K_3PO_4

$$NaCN + H_2O \rightleftharpoons NaOH + HCN$$

$$Na^+ + CN^- + H_2O \rightleftharpoons Na^+ + OH^- + HCN$$

$$CN^- + H_2O \rightleftharpoons HCN + OH^-$$

(i) In this type of salt hydrolysis anion reacts with water therefore called as anionic hydrolysis.

(ii) Solution is basic in nature as $[OH^-]$ increases.

(iii) pH of the solution is greater than 7.

(iv) Relation between K_h, K_w, K_a

$$CN^- + H_2O \rightleftharpoons HCN + OH^-$$

$$K_h = \frac{[HCN][OH^-]}{[CN^-]} \qquad \text{..... (i)}$$

$$\frac{[HCN][OH^-]}{[CN^-]} \times \frac{[CN^-][H^+]}{[HCN]} = [H^+][OH^-]$$

$$\boxed{K_h = \frac{K_w}{K_a}}$$

(v) Degree of hydrolysis :

$CN^- + H_2O \rightleftharpoons$	HCN	+	OH^-	
C	0		0	Initial concentration at equlibrium
C – Ch	Ch		Ch	

$$K_h = \frac{[HCN][OH^-]}{[CN^-]}$$

$$K_h = \frac{Ch^2}{(1-h)}$$

Since $h <<<< 1$, therefore $(1 - h) \approx 1$

$\therefore$ $K_h = Ch^2$

$$h^2 = \frac{K_h}{C} \qquad \Rightarrow \qquad \boxed{h = \sqrt{\frac{K_h}{C}}}$$

$$\boxed{h = \sqrt{\frac{K_w}{K_a \times C}}}$$

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(vi) pH of the solution

$[OH^-] = Ch$

$$\left[OH^-\right] = \sqrt{\frac{K_w \times C}{K_a}}$$

$$\left[H^+\right] = \frac{K_w}{\sqrt{\frac{K_w \times C}{K_a}}} \Rightarrow \boxed{\left[H^+\right] = \sqrt{\frac{K_w \times K_a}{C}}}$$

On taking – log on both sides

$$pH = -\frac{1}{2}[\log K_w + \log K_a - \log C]$$

$$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$$

Ex.22 Calculate the pH and degree of hydrolysis of 0.01 M solution of NaCN, K_a for HCN is 6.2×10^{-12}.

Sol. NaCN is a salt of strong base NaOH and weak acid HCN. Na^+ does not react with water whereas CN^- reacts with water as here under

$$CN^- + H_2O \rightleftharpoons HCN + OH^-$$

$$K_h = \frac{[HCN][OH^-]}{[CN^-]} = \frac{K_w}{K_a} = \frac{10^{-14}}{6.2\times 10^{-12}} = 1.6 \times 10^{-3}$$

Let, x moles of salt undergo hydrolysis then concentrations of various species would be

$[CN^-] = (0.01 - x) \approx 0.01$, $[HCN] = x$

$[OH^-] = x$

$\therefore \quad K_h = \frac{x.x}{0.01} = 1.6 \times 10^{-3}$

$\therefore \quad x^2 = 1.6 \times 10^{-5}$

$\therefore \quad x = 4 \times 10^{-3}$

$[OH^-] = x = 4 \times 10^{-3}$ M

$$[H_3O^+] = \frac{K_W}{[OH^-]} = \frac{10^{-14}}{4\times 10^{-3}} = 0.25 \times 10^{-11}$$

$pH = -\log(0.25 \times 10^{-11}) = 11.6020$

Degree of hydrolysis $= \frac{x}{0.01} = \frac{4\times 10^{-3}}{0.01} = 4 \times 10^{-1}$

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Ex.23. Calculate for 0.01 N solution of sodium acetate -

(i) Hydrolysis constant

(ii) Degree of hydrolysis

(iii) pH

Given K_a of CH_3COOH = 1.9×10^{-5}.

Sol. For $CH_3COONa + H_2O \rightleftharpoons CH_3COOH + NaOH$

Initial	C	0	0
After	C(1–h)	Ch	Ch

(i) $K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{1.9 \times 10^{-5}} = 5.26 \times 10^{-10}$

(ii) $h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{5.26 \times 10^{-10}}{0.01}} = 2.29 \times 10^{-4}\,M$

(iii) $[OH^-]$ from NaOH, a strong base = $Ch = 0.01 \times 2.29 \times 10^{-4} = 2.29 \times 10^{-6}$ M

pOH = 5.64

$\therefore$ pH = 14 – 5.64 = 8.36

5.2.3 Hydrolysis of (WA - WB) type salt :

Ex. NH_4CN, $CaCO_3$, $(NH_4)_2\,CO_3$, $ZnHPO_3$

$NH_4CN + H_2O \rightleftharpoons NH_4OH + HCN$

$NH_4^+ + CN^- + H_2O \rightleftharpoons NH_4OH + HCN$

Solution is almost neutral but it may be acidic or basic depending upon the nature of acid & base & pH of the solution is near to 7.

For WA - WB types of salt :

	$K_a > K_b$	$K_b > K_a$	$K_a = K_b$
1. Hydrolysis	Cationic-anionic	Anionic-cationic	Neutral hydrolysis
2. Nature	Acidic	Basic	Neutral
3. pH	pH < 7	pH > 7	pH = 7

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(i) Relation between K_h, K_w, K_a & K_b

$$NH_4^+ + CN^- + H_2O \rightleftharpoons NH_4OH + HCN$$

$$K_h = \frac{[NH_4OH][HCN]}{[NH_4^+][CN^-]} \quad \text{..... (i)}$$

$$\frac{[NH_4OH][HCN]}{[NH_4^+][CN^-]} \times \frac{[NH_4^+][OH^-]}{[NH_4OH]} \times \frac{[H^+][CN^-]}{[HCN]} = [H^+]\,[OH^-]$$

$$K_h \times K_b \times K_a = K_w$$

$$\boxed{K_h = \frac{K_w}{K_a \times K_b}}$$

(ii) Degree of Hydrolysis :

NH_4^+ +	CN^- + H_2O $\rightleftharpoons$	NH_4OH +	HCN	
C	C	0	0	Initial concentration
C – Ch	C – Ch	Ch	Ch	at equilibrium

$$K_h = \frac{[NH_4OH][HCN]}{[NH_4^+][CN^-]}$$

Since $h <<<< 1$

Then $(1 - h) \approx 1$

$\therefore$ $\boxed{K_h = h^2}$ or $h^2 = \frac{K_W}{K_a \times K_b}$

$$h = \sqrt{\frac{K_W}{K_a \times K_b}} \quad \text{..... (v)}$$

(iii) pH of the solution

From eq. (iii)

$$K_a = \frac{[H^+][CN^-]}{[HCN]}$$

$$[H^+] = \frac{K_a \times [HCN]}{[CN^-]}$$

$$[H^+] = \frac{K_a \times Ch}{C - Ch} = \frac{K_a \times h}{1 - h}$$

Since $h <<<< 1$, $(1 - h) \approx 1$

$[H^+] = K_a \times h$ [Now put the value of h from eq. (5)]

$$= K_a \times \sqrt{\frac{K_w}{K_a \times K_b}}$$

$$\boxed{[H^+] = \sqrt{\frac{K_w \times K_a}{K_b}}}$$

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On taking – log on both sides

$$-\log [H^+] = -\log\left(\frac{K_w \times K_a}{K_b}\right)^{1/2}$$

$$pH = -\frac{1}{2}\ [\log K_w + \log K_a - \log K_b]$$

$$pH = 7 + \frac{1}{2}\ pK_a - \frac{1}{2}\ pK_b$$

Note : Degree of hydrolysis of [WA – WB] type salt does not depend on the concentration of salt.

Ex.24 Salt of weak acid and weak base

(i) Calculate pH of the mixture (25 mL of 0.1 M NH_4OH + 25 mL of 0.1 M CH_3COOH).

Given that K_a : 1.8×10^{-5}, and $K_b = 1.8 \times 10^{-5}$

Sol.

	NH_4OH	+ CH_3COOH	→ CH_3COONH_4	+ H_2O
Initial milli moles	25×0.1	25×0.1	0	0
	$= 2.5$	$= 2.5$	–	
Final milli moles	0	0	2.5	2.5

As salt is formed (salt of weak acid and weak base) and pH will be decided by salt hydrolysis

$$pH = \frac{pK_w + pK_a - pK_b}{2} = \frac{1}{2}(-\log 10^{-14} - \log 1.8 \times 10^{-5} + \log 1.8 \times 10^{-5}) = 7$$

Ex.25 In the following which one has highest / maximum degree of hydrolysis.

(1) 0.01 M – NH_4Cl ***(2) 0.1 M –NH_4Cl***

(3) 0.001 M –NH_4Cl ***(4) Same***

Sol. [3]

$\left(h = \sqrt{\frac{K_h}{C}}\right.$ if C decreases, h increases$\left.\right)$

Ex.26 In the following which one has lowest value of degree of hydrolysis.

(1) 0.01 M – CH_3COONH_4 ***(2) 0.1 M – CH_3COONH_4***

(3) 0.001 M – CH_3COONH_4 ***(4) Same***

Sol. [4]

Ex.27 ***Find out the concentration of $[H^+]$ in 0.1M CH_3COONa solution ($K_a = 10^{-5}$)***

Sol. *Salt is [WA – SB] type*

$$\therefore \quad [H^+] = \sqrt{\frac{K_w \times K_a}{C}} = \sqrt{\frac{10^{-14} \times 10^{-5}}{10^{-1}}} = \sqrt{10^{-19} \times 10^{+1}} = \sqrt{10^{-18}} = 10^{-9}$$

Ex.28 Calculate the degree of hydrolysis of a mixture containing 0.1N NH_4OH and 0.1N HCN

$K_a = 10^{-5}$ & $K_b = 10^{-5}$

Sol. *Salt is [WA – WB]*

$$h = \sqrt{\frac{K_w}{K_a \times K_b}} = \sqrt{\frac{10^{-14}}{10^{-5} \times 10^{-5}}}$$

$$= \sqrt{10^{-14} \times 10^{+10}} = \sqrt{10^{-4}} = 10^{-2}$$

5.2.4 Hydrolysis of [SA – SB] type salt :

Ex. NaCl, $BaCl_2$, Na_2SO_4, $KClO_4$ etc.

$$NaCl + H_2O \rightleftharpoons NaOH + HCl$$

$$Na^+ + Cl^- + H_2O \rightleftharpoons Na^+ + OH^- + H^+ + Cl^-$$

$$H_2O \rightleftharpoons H^+ + OH^- \text{ (It is not salt hydrolysis)}$$

(1) Hydrolysis of salt of [SA – SB] is not possible

(2) Solution is neutral in nature (pH = pOH = 7)

(3) pH of the solution is 7

5.2.5 Hydrolysis of Amphiprotic Anion :

$NaHCO_3$, NaHS, etc., can undergo ionisation to from H^+ ion and can undergo hydrolysis to from OH^- (Na^+ ion is not hydrolysed)

(a) (i) $HCO_3^- + H_2O \overset{\text{ionisation}}{\rightleftharpoons} CO_3^{2-} + H_3O^+$ (acid)

(ii) $HCO_3^- + H_2O \overset{\text{hydrolysis}}{\rightleftharpoons} H_2CO_3 + OH^-$ (base)

$$pH(HCO_3^-) = \left(\frac{pK_{a_1} + pK_{a_2}}{2}\right)$$

(b) Similarly for $H_2PO_4^-$ and HPO_4^{2-} amphiprotic anions.

$$pH_{(H_2PO_4^-)} = \left(\frac{pK_{a_1} + pK_{a_2}}{2}\right) \quad \text{and} \quad pH_{(HPO_4^{2-})} = \left(\frac{pK_{a_2} + pK_{a_3}}{2}\right)$$

$$\underset{CM}{NaHCO_3} \longrightarrow \underset{C}{Na^+} + \underset{C}{HCO_3^-}$$

$$HCO_3^- + H_2O \overset{Ka/Ka_1}{\rightleftharpoons} H_2CO_3 + OH^-$$

$$HCO_3^- + H_2O \overset{Ka_2}{\rightleftharpoons} CO_3^{-2} + H_3O^+$$

$\because$ H^+ and OH^- also react

$\therefore$ We can safely assume that both reactions have nearly same degree of dissociation

$$\therefore [H_2CO_3] \approx [CO_3^{-2}] \qquad .....(1)$$

$$\frac{Kw}{Ka_1} = \frac{[H_2CO_3][OH^-]}{[HCO_3^-]} \Rightarrow \frac{1}{Ka_1} = \frac{[H_2CO_3]}{[H^+][HCO_3^-]} \qquad ....(2)$$

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$$\frac{[CO_3^{-2}][H^+]}{[HCO_3^-]} = Ka_2 \qquad ....(3)$$

Divide (2) by (3)

$$[H^+] = \sqrt{Ka_1 Ka_2} \Rightarrow pH = \frac{pKa_1 + pKa_2}{2}$$

Ex.29 Calculate the pH of 0.5 M Na_3PO_4 in aqueous solution ?

$$PO_4^{3-} + H_2O \rightleftharpoons HPO_4^{2-} + OH^- \; ; \; K_b(PO_4^{-3}) = 2.4 \times 10^{-2}$$

Sol. *HPO_4^{2-} and PO_4^{-3} are conjugate acid and base so $K_a \times K_b = 10^{-14}$*

$$K_a(HPO_4^{2-}) = \frac{10^{-14}}{2.4 \times 10^{-2}} = 4.17 \times 10^{-13}$$

$$pK_a = -\log K_a = 12.38$$

$$or \quad pH = 7 + \frac{1}{2} pK_a + \frac{1}{2} \log C$$

$$pH = 13.04$$

6 BUFFER SOLUTIONS

A solution that resists change in pH value upon addition of small amount of strong acid or base or when solution is diluted is called buffer solution.

The capacity of a solution to resist alteration in its pH value is known as buffer capacity and the mechanism of buffer solution is called buffer action.

6.1 Types of buffer solutions

(A) Simple buffer solution (B) Mixed buffer solution

6.2 SIMPLE BUFFER SOLUTION :

A salt of weak acid and weak base in water e.g. CH_3COONH_4, $HCOONH_4$, AgCN, NH_4CN.

Buffer action of simple buffer solution

Consider a simple buffer solution of CH_3COONH_4 ,since it is a salt will dissociated completely.

$$CH_3COONH_4 \longrightarrow CH_3COO^- + NH_4^+$$

If a strong acid such as HCl is added then

$$HCl \longrightarrow H^+ + Cl^-$$

The H^+ ions from the added acid (HCl) combine with CH_3COO^- ions to form CH_3COOH, which is a weak acid so will not further ionized.

Thus there is no rise in H^+ ion concentration and the pH remains constant.

$$CH_3COO^- + H^+ \rightleftharpoons CH_3COOH \text{ (Weak acid)}$$

- If a strong base is added as NaOH

$$NaOH \longrightarrow Na^+ + OH^-$$

$$NH_4^+ + OH^- \rightleftharpoons NH_4(OH) \text{ (Weak base)}$$

Thus change in OH^- ion concentration is resisted by NH_4^+ ions by forming NH_4OH which is a weak base. So it will not further ionized and pH remains constant.

pH of a simple buffer solution :-

$$\boxed{pH = 7 + \frac{1}{2} pk_a - \frac{1}{2} pk_b}$$

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6.3 MIXED BUFFER SOLUTIONS :

6.3.1 Acidic buffer solution :

An acidic buffer solution consists of solution of a weak acid and its salt with strong base. The best known example is a mixture of solution of acetic acid and its salt with strong base (CH_3COONa). Other example :

$HCN + KCN, (H_2CO_3 + NaHCO_3) \longrightarrow$ blood

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ (Weakly ionised)

$CH_3COONa \longrightarrow CH_3COO^- + Na^+$ (Highly ionised)

When a few drops of an acid (HCl) are added to it, the H^+ ions from the added acid (HCl) combine with the CH_3COO^- ions to form CH_3COOH. Thus there is no rise in H^+ ion concentration and the pH of solution remains constant. On the other hand, when a few drops of base(NaOH) are added, the OH^- of the added base reacts with acetic acid to form unionise water and acetate ions.

$CH_3COOH + OH^- \rightleftharpoons H_2O + CH_3COO^-$

Thus there is no increase in OH^- ion concentration and hence the pH of the solution remains constant.

pH of a acidic buffer solution (Henderson equation) :

Consider a buffer mixture (acidic buffer)

$HA + NaA$ $(CH_3COOH + CH_3COONa)$

where $A = CH_3COO$, $A^- = CH_3COO^-$

$HA \rightleftharpoons H^+ + A^-$

$NaA \longrightarrow Na^+ + A^-$

Applying law of mass action to dissociation equilibrium of HA

$$K_a = \frac{[H^+][A^-]}{[HA]} \; ; \; \text{so} \; [H^+] = \frac{K_a[HA]}{[A^-]}$$

taking log, $$\log [H^+] = \log K_a + \log \frac{[HA]}{[A^-]}$$

$$-\log [H^+] = -\log K_a - \log \frac{[HA]}{[A^-]}$$

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

$[A^-]$ = Initial concentration of salt as it is mainly comes from salt.

[HA] = Initial concentration of the acid.

$$pH = pK_a + \log \frac{[Salt]}{[Acid]}$$ (it is known as Henderson-Hasselbalch equation.)

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Note : A solution can act as buffer only if ratio of concentration of salt to acid is between 0.1 to 10.

CH_3COOH :	CH_3COONa	
1	10	$pH = pK_a + 1$
10	1	$pH = pK_a - 1$

Thus pH range of an acidic buffer solution is $(pK_a + 1)$ to $(pK_a - 1)$

$pH \text{ range} = pK_a \pm 1$

Maximum buffer action will be only when ratio of concentration of acid and salt is 1. So for maximum buffer action,

$pH = pK_a$

Ex.30 How much volume of 0.2 M solution of acetic acid should be added to 100 mL of 0.2 M solution of sodium acetate to prepare a buffer solution of pH = 6.00 ? (pK_a for acetic acid is 4.74)

Sol. $pH = pK_a + \log \frac{[Salt]}{[Acid]}$

$\log \frac{[Salt]}{[Acid]} = pH - pK_a = 6.00 - 4.74 = 1.26 \quad \therefore \quad \frac{[Salt]}{[Acid]} = 18.2$

Moles of CH_3COONa in solution $\frac{100 \times 0.2}{1000} = 0.02$

Let, volume of 0.2 acetic acid added = V mL

$\therefore$ Moles of acetic acid $= \frac{V \times 0.2}{1000}$

$\therefore \quad \frac{0.02}{V \times \frac{0.2}{1000}} = 18.2$

$\therefore \quad V = 5.49$ mL

Ex.31 Calculate the pH after the addition of 80 mL and 100 mL respectively of 0.1 N NaOH to 100 mL, 0.1 N CH_3COOH. (Given pK_a for CH_3COOH = 4.74)

Sol. *If 80 mL of 0.1 N NaOH is added to 100 mL of 0.1 N CH_3COOH, acidic buffer will form as*

	H_3CCOOH +	$NaOH$ ⟶	$H_3CCOONa$ +	H_2O
Initial	*0.01 eq.*	*0.008 eq.*	*0*	*0*
Final	*0.002 eq.*	*0*	*0.008 eq.*	

$pH = pK_a + \log \frac{[CH_3COO^-]}{[CH_3COOH]} = 4.74 + \log \frac{0.008}{0.002} = 5.342$

If 100 mL of 0.1 N NaOH is added is added to 100 mL of 0.1 N CH_3COOH, complete neutralization takes place and the concentration of $H_3CCOONa = \frac{0.1}{2} M = 0.05\ M$

Now, $pH = 7 + \frac{1}{2} pK_a + \frac{1}{2} \log C = 8.72$

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Ex.32 Calculate the pH of a solution when 0.20 moles of HCl is added to one litre solution containing -

(a) 1 M each of acetic acid and acetate ion ?

(b) 0.1 M each of acetic acid and acetate ion ?

Given K_a for acetic acid is 1.8×10^{-5}.

Sol. ***(a)*** *Initially [Acetic acid] =1 M*

[Acetate] = 1 M

Now 0.2 moles of HCl are added to it.

	HCl	+	CH_3COO^-	$\rightarrow$	CH_3COOH	+	Cl^-
Mole before reaction	*0.2*		*1*		*1*		*0*
Mole after reaction	*0*		*0.8*		*1.2*		*0.2*

$\therefore$ *New* $[CH_3COOH] = 1.2$; $[CH_3COO^-] = 0.8$

$\therefore$ $pH = pk_a + \log \frac{[\text{conjugate}]}{[\text{acid}]}$

$\therefore$ $pH = -\log 1.8 \times 10^{-5} + \log \frac{0.8}{1.2} = 4.5686$

(b) *In II case initially [Acetic acid] = 0.1 M*

[Acetate] = 0.1 M

Now 0.2 mole of HCl are added to it

	HCl	+	CH_3COO^-	$\rightarrow$	CH_3COOH	+	Cl^-
Mole before reaction	*0.2*		*0.1*		*0.1*		*0*
Mole after reaction	*0.1*		*0*		*0.2*		*0.1*

$\therefore$ $[H^+]$ *from free HCl = 0.1 M*

$\therefore$ *pH = 1*

Note: *CH_3COOH no doubt gives H^+ but being weak acid as well as in presence of HCl does not dissociate appreciably and thus, H^+ from CH_3COOH may be neglected.*

6.3.2 Basic buffer solution :

A basic buffer solution consists of a mixture of a weak base and its salt with strong acid. The best known example is a mixture of NH_4OH and NH_4Cl.

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$ (Weakly ionised)

$NH_4Cl \rightarrow NH_4^+ + Cl^-$ (Highly ionised)

When a few drops of a base (NaOH) are added, the OH^- ions from NaOH combine with NH_4^+ ions to form feebly ionised NH_4OH thus there is no rise in the concentration of OH^- ions and hence the pH value remains constant.

$NH_4^+ + OH^- \rightleftharpoons NH_4OH$

If a few drops of a acid (HCl) are added the H^+ from acid combine with NH_4OH to form H_2O and NH_4^+ ions.

$NH_4OH + H^+ \rightleftharpoons NH_4^+ + H_2O$

Thus the addition of acid does not increase the H^+ ion concentration and hence pH remains unchanged.

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- **pH of basic buffer solution :**

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

$$NH_4Cl \rightarrow NH_4^+ + Cl^-$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]}$$

$$[OH^-] = \frac{K_b[NH_4OH]}{[NH_4^+]} = \frac{K_b[Base]}{[Salt]}$$

(NH_4^+ mainly comes from salt)

taking –log on both side

$$-\log OH^- = -\log \frac{K_b[Base]}{[Salt]} \Rightarrow pOH = -\log K_b - \log \frac{[Base]}{[Salt]}$$

$$pOH = pK_b + \log \frac{[Salt]}{[Base]} \Rightarrow pH = 14 - pOH$$

❖ **pOH range :**

A solution can act as buffer solution only if ratio of concentration of salt to base is from 0.1 to 10.

NH_4OH	:	NH_4Cl	
1		10	$pOH = pK_b + 1$
10		1	$pOH = pK_b - 1$

So pOH range is $pK_b \pm 1$

❖ **Condition for maximum buffer action :**

$[NH_4OH]$	:	$[NH_4Cl]$
1		1

$$pOH = pK_b + \log \frac{1}{1}$$

$$pOH = pK_b \quad \text{and} \quad pH = 14 - pK_b$$

Maximum buffer action because pH remains constant.

Ex.33 An organic base B has K_b value equal to 1×10^{-8}. In what amounts should 0.01 M HCl and 0.01 M solution of B be mixed to prepare 1 L of a buffer solution having pH = 7.0 ?

Sol. $B + H_2O \rightleftharpoons BH^+ + OH^-$

$$K_b = \frac{[BH^+][OH^-]}{[B]} = 1 \times 10^{-8}$$

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$$pOH = pK_b + \log \frac{[BH^+]}{[B]}$$

$\Rightarrow \quad 7 = -\log(10^{-8}) + \log\frac{[BH^+]}{[B]} \Rightarrow 7 = 8 + \log\frac{[BH^+]}{[B]}$

$$\log \frac{[BH^+]}{[B]} = -1$$

$\therefore \quad \frac{[BH^+]}{[B]} = 10^{-1} = 0.1$

Let, volume of HCl taken = xL

$\therefore$ Volume of base taken = (1 – x) L

After the reaction, millimole of BH^+ formed = 0.01 × (x)

Millimoles of base left = 0.01 (1 – 2x)

$\therefore \quad \frac{[BH^+]}{[B]} = \frac{x}{[1-2x]} = 0.1$

$\therefore$ x = 0.083 L = Volume of HCl

$\therefore$ Volume of base = 0.917 L

Ex.34 Which of the following buffers containing NH_4OH and NH_4Cl show the lowest pH value?

	conc. of NH_4OH (mol L^{-1})	***conc. of NH_4Cl (mol L^{-1})***
(A)	***0.50***	***0.50***
(B)	***0.10***	***0.50***
(C)	***0.50***	***1.50***
(D)	***0.50***	***0.10***

Sol. (B) $pOH = pk_b + \log \frac{[salt]}{[base]}$ for $NH_4Cl = 0.5$ and $NH_4OH = 0.1$

pOH will be maximum and so pH will be minimum.

Ex.35 A solution of weak base BOH was titrated with 0.1 N HCl. The pH of the solution was found to be 10.04 and 9.14 after the addition of 5 mL and 20 mL of the acid respectively. Find the dissociation constant of the base.

Sol. Case I :

	BOH	+	HCl	⟶	BCl	+	H_2O
Millimole before reac.	a		0.1× 5=0.5		0		0
Millimole after reac.	(a–0.5)		0		0.5		0.5

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$\therefore$ $pOH = -\log K_b + \log \frac{[BCl]}{[BOH]}$ .....(i)

$\because$ pH = 10.04 so pOH = 3.96

$\therefore$ $3.96 = -\log K_b + \log \frac{0.5}{(a-0.5)}$ .....(ii)

Case II :

	BOH	+ HCl	⟶	BCl	+ H_2O
Millimole before reac.	a	$0.1\times 20 = 2$		0	0
Millimole after reac.	(a–2)	0		2	2

$\therefore$ $pOH = -\log K_b + \log \frac{[BCl]}{[BOH]}$ .....(iii)

$\because$ pH = 9.14 $\therefore$ pOH = 4.86

$\therefore$ $4.86 = -\log K_b + \log \frac{2}{(a-2)}$ .....(iv)

Solving Eqs. (ii) and (iv), $K_b = 1.81 \times 10^{-5}$

7. INDICATORS

The stage of titration when complete reaction occur between the solution is called **equivalent point.** The stage of titration when sudden change in colour of solution is observed is called **end point.** A perfect indicator response sudden colour change exactly on completion of reaction. An **indicator** is a substance which response sudden change in colour of solution at the end point or neutral point of the acid-base titration. **At end point** $N_1V_1 = N_2V_2$

(i) The indicators in acid-base titration changes colour on changing the pH of solution.

(ii) All the acid-base indicators are either weak organic acid or base and having different colour for unionized and ionised form.

(iii) A mixture of two colour is recognized in a single colour if the conc. of one is 10 times or more than that of others. (This 10 time is flexible)

$$HA \rightleftharpoons A^- + H^+$$

colour X colour Y

Diss. const. or (Ionisation const) $= K_a = K_{in} = \frac{[H^+][A]}{[HA]}$

$$pH = pK_{in} + \log\frac{[A^-]}{[HA]}$$

(a) The solution will appear only of colour Y, if $\frac{[A^-]}{[HA]} \geq 10 \Rightarrow pH \geq (pK + 1)$

(b) The solution will appear only of colour X, if $\frac{[A^-]}{[HA]} \leq \frac{1}{10} \Rightarrow pH \leq (pK - 1)$

pH of solution below and above which solution appears in a single colour is called pH range of indicator.

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Indicator	pH range	Colour change	pK_a
Methyl orange	3.2 – 4.5	Pink to yellow	3.7
Methyl red	4.4 – 6.5	Red to yellow	5.1
Litmus	5.5 – 7.5	Red to blue	7.0
Phenol red	6.8 – 8.4	Yellow to red	7.8
Phenolpthalein	8.3 – 10.5	Colourless to pink	9.6

Ex.36 The disso. const. of a basic indicator is 2×10^{-7}. Calculate its pH range.

Sol. 5.7 – 7.7 = pOH $\quad\therefore\quad$ pH = 6.3 – 8.3

Ex.37 The pH range of an acidic indicator HIn is 4.0 - 5.2 . Calculate dissociation constant. Also calculate $\frac{In^-}{HIn}$ for the appearence of solution in single colour.

Sol. Diss. constant = 2.5×10^{-5} , 4

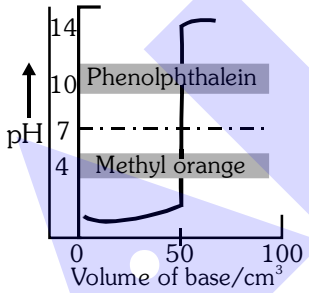
9.1 TITRATION OF STRONG ACID AGAINST STRONG ALKALI :

The graph (A) shows how pH changes during the titration of 50 cm^3 of 0.1 M HCl with 0.1 M NaOH.

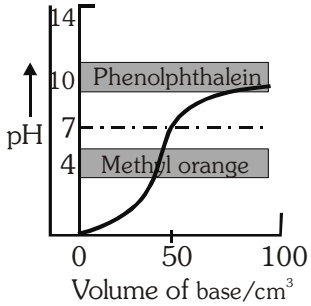
$$NaOH\ (aq) + HCl\ (aq) \longrightarrow NaCl\ (aq) + H_2O\ (\ell)$$

The pH of 0.1 M solution of HCl in the beginning would be 1. As alkali is added, the pH changes slowly in the beginning. However, at the equivalence point pH changes rapidly from about 3.5 to 10. It can be shown by simple calculations that pH of the solution is 3.7 when 49.8 cm^3 of 0.1 M NaOH solution have been added. The pH suddenly changes to 10 after addition of 50.1 cm^3 of the NaOH solution. Thus, any indicator having pH range between 3.5 to 10 will identify the equivalence point. This means that any one of phenolphthalein, methyl orange or bromothymol blue could be used as an indicator.

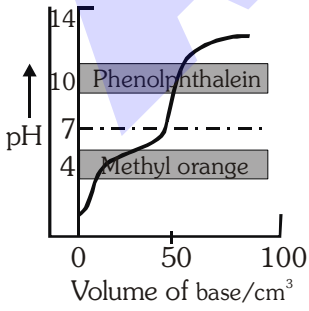
(A)



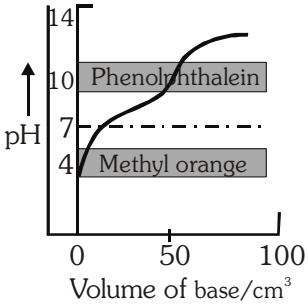
(B)



(C)



(D)



Titration curves : (A) strong base with strong acid ; (B) weak base with strong acid ; (C) strong base with weak acid ; (D) weak base with weak acid.

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7.2 TITRATION OF STRONG ACID AGAINST WEAK ALKALI :

The graph (B) shows how pH changes during titration of 50 cm^3 of 0.1 M HCl with 0.1 M NH_3.

$$HCl\ (aq) + NH_4OH\ (aq) \longrightarrow NH_4Cl\ (aq) + H_2O\ (\ell)$$

In this case, the pH changes rapidly from 3.5 to 7.0 at the equivalence point. Methyl orange, methyl red and bromocresol green are suitable indicators for this type of titration. Phenolphthalein is unsuitable because its pH range lies outside the vertical portion of the curve.

7.3 TITRATION OF WEAK ACID AGAINST STRONG BASE :

The graph (C) shows how pH changes during titration of 50 cm^3 of 0.1 M CH_3COOH with 0.1 M NaOH.

$$CH_3COOH\ (aq) + NaOH\ (aq) \longrightarrow CH_3COONa\ (aq) + H_2O\ (\ell)$$

The vertical portion of this titration curve lies between pH range 7 to 10.6. Phenolphthalein is suitable indicator for this titration. Methyl orange is not suitable for this titration because its pH range lies on the flat portion of the curve.

7.4 TITRATION OF WEAK ACID AGAINST WEAK BASE :

The graph (D) represents the titration curve obtained for titration of 50 cm^3 of 0.1 M CH_3COOH with 0.1 M NH_3.

$$CH_3COOH\ (aq) + NH_4OH\ (aq) \longrightarrow CH_3COONH_4\ (aq) + H_2O\ (\ell)$$

For this type of titration there is no sharp increase in pH at the equivalence point. No indicator is suitable for this type of titration.

Ex.38 Bromophenol blue is an indicator with a value of $K_a = 6.84 \times 10^{-6}$. At what pH it will work as an indicator? Also report the % of this indicator in its basic form at a pH of 5.84.

Sol. $HBPh \rightleftharpoons H^+ + BPh^-$

$K_a = \dfrac{[H^+][BPh^-]}{[HBPh]}$, when $BPh^- = HBPh$, indicator will work. Thus

$[H^+] = 6.84 \times 10^{-6}$

$\therefore$ pH = 5.165

Also if pH = 5.84

or $[H^+] = 1.44 \times 10^{-6}$, then

$K_a = \dfrac{[H^+][BPh^-]}{[HBPh]}$ or $6.84 \times 10^{-6} = \dfrac{1.44 \times 10^{-6}.C\alpha}{C(1-\alpha)}$ or $\alpha = 0.83$ or 83 %

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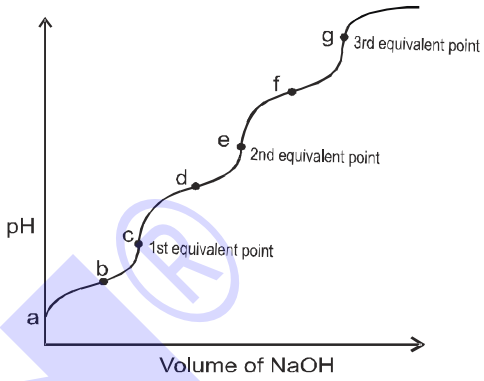
7.5 Titration of Poly protic acid v/s S.B. :

50 mL of 0.1 M H_3PO_4 agent 0.1 M NaOH. Calculate pH when vol. of NaOH added is

(a) 0 mL (b) 25 mL (c) 50 mL (d) 75 mL (e) 100 mL

(f) 125 mL (g) 150 mL (h) 200 mL (i) 90 mL

$pK_{a1} = 3$ $pK_{a2} = 7$, $pK_{a3} = 11$



(a) $pH = 1/2\ (pK_{a1} - \log C) = 1/2\ (3 + 1) = 2$

(b) $H_3PO_4 + NaOH \rightleftharpoons NaH_2PO_4 + H_2O$

t = 0 5 Mmol 2.5 Mmol 0 –

2.5 M mol 0 2.5 M mol

$pH = pK_{a1} + \log 1/1 = pK_{a1} = 3$

(c) Solution of $H_2PO_4^-$ amphoteric species

$$pH = \frac{pK_{a1} + pK_{a2}}{2} = \frac{3+7}{2} = 5$$

(d) $NaH_2PO_4 + NaOH \longrightarrow Na_2HPO_4 + H_2O$

t = 0 5 M mol 2.5 M mol 0 –

2.5 M mol 2.5 M mol –

$pH = \log 1/1 + pK_{a2} = 7$

(e) HPO_4^{2-} solution (amphoteric species)

$$pH = \frac{pK_{a2} + pK_{a3}}{2} = 9$$

(f) $Na_2HPO_4 + NaOH \longrightarrow Na_3PO_4 + H_2O$

5 2.5 0 –

2.5 0 2.5 –

$pH = pK_{a3} + \log 1/1 = 0$

(g) 3rd eq. pt Na_3PO_4 solution

$[Na_3PO_4] = 5/200 = 1/40$

$pH = 1/2\{pK_w + pK_{a3} + \log C\} = 1/2\ (14 + 11 - 2 + 0.4) = 11.7$

(h) 200 mL

hydrolysis of Na_3PO_4 can be neglected in presence of NaOH

[NaOH] = 5/250 = 1/50 pOH = 1.7

pH = 12.3

(i) 90 mL

$NaH_2PO_4 + NaOH \longrightarrow Na_2HPO_4 + H_2O$

5 4 0

1 0 4

$pH = pK_{a2} + \log 4 = 7.6$

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10. SOLUBILITY (s) & SOLUBILITY PRODUCT (K_{sp})

10.1 SOLUBILITY :

At constant temperature, the maximum number of moles of solute which can be dissolved in a solvent to obtain 1 litre of saturated solution is called solubility.

Solubility depends on the following –

(i) Temperature

(ii) Presence of common ion

(iii) Nature of solvent

10.2 SOLUBILITY PRODUCT(K_{sp}) :

When a sparingly soluble salt such as AgCl is put into water, a very small amount of AgCl dissolves in water and most of the salt remains undissolved in its saturated solution.

- A solution which remains in contact with undissolved solute is said to be saturated.
- The salt AgCl is an elecrolyte, its dissociation occurs in solution. Hence, the quantity of AgCl that dissolves in water dissociates into Ag^+ and Cl^- ions. Thus, in the saturated solution of AgCl an equilibrium exists between undissolved solid AgCl and its ions, Ag^+ and Cl^- ions.

$$AgCl_{(s)} \underset{\text{Precipitation}}{\overset{\text{Dissolution}}{\rightleftharpoons}} Ag^+_{(aq)} + Cl^-_{(aq)}$$

according to law of mass action

$$K = \frac{[Ag^+].[Cl^-]}{[AgCl]}$$

Since, the concentration of undissolved solid AgCl is constant. Thus, the product K.[AgCl] gives another constant which is designated as K_{sp}

So, $K.[AgCl] = [Ag^+].[Cl^-]$

$\therefore$ $K_{sp} = [Ag^+] . [Cl^-]$

- **K_{sp} for $CaCl_2$** $CaCl_2(s) \rightleftharpoons Ca^{+2}(aq) + 2Cl^-(aq)$

  Solubility product in terms of concentration of ions $K_{sp} = [Ca^{+2}] [Cl^-]^2$

- **K_{sp} for $AlCl_3$** $AlCl_3(s) \rightleftharpoons Al^{+3}(aq) + 3Cl^-(aq)$

  Solubility product in terms of concentration of ions $K_{sp} = [Al^{+3}] [Cl^-]^3$

- **General form** $A_xB_y(s) \rightleftharpoons xA^{+y}(aq) + yB^{-x}(aq)$

  $K_{sp} = [A^{+y}]^x [B^{-x}]^y$

Thus, solubility product is defined as the product of concentrations of the ions raised to a power equal to the number of ions given by the dissociation of electrolyte at a given temperature when the solution is saturated.

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10.3 APPLICATION OF SOLUBILITY PRODUCT (K_{sp}) :

10.3.1 To find out the solubility (S) :

(i) K_{sp} of AB (Mono-mono, di-di, tri-tri valency) type salt –

Ex. NaCl, $BaSO_4$, CH_3COONa, $CaCO_3$, NaCN, KCN, NH_4CN, NH_4Cl etc.

$$AB(s) \rightleftharpoons A^+(aq) + B^-(aq)$$

a	0	0
(a–s)	s	s

$K_{sp} = [A^+]\,[B^-]$

$K_{sp} = s^2$ or $s = \sqrt{K_{sp}}$

(ii) K_{sp} of AB_2 or A_2B (Mono-di or di-mono valency) type salt –

Ex. $CaCl_2$, $CaBr_2$, K_2S, $(NH_4)_2SO_4$, K_2SO_4, K_2CO_3 etc.

$$AB_2(s) \rightleftharpoons A^{+2}(aq) + 2B^-(aq)$$

a	0	0
a – s	s	2s

$K_{sp} = [A^{+2}]\,[B^-]^2$

$K_{sp} = s \times (2s)^2 = s \times 4s^2 = 4s^3$

$$s = \left(\frac{K_{sp}}{4}\right)^{1/3}$$

(iii) General form :

$$A_xB_y(aq) \rightleftharpoons xA^{+y}(aq) + yB^{-x}(aq)$$

a	0	0
a–s	xs	ys

$K_{sp} = [A^{+y}]^x \,.\, [B^{-x}]^y$

$K_{sp} = (xs)^x .\, (ys)^y$

$$\boxed{K_{sp} = x^x . y^y . s^{(x+y)}}$$

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10.4 COMMON ION EFFECT ON SOLUBILITY :

Solubility of substances always decreases in the presence of common ion. According to Le-Chatelier's principle, on increasing common ion concentration equilibrium shifts in backward direction until the equilibrium is reestablished so, the solubility of substances decreases.

Ex. Find out the solubility of AgCl in water and in the presence of CM – NaCl solution?

$$AgCl \rightleftharpoons Ag^+ + Cl^- \qquad CM.NaCl$$

$$S \qquad S \qquad \text{(Let solubility of AgCl is S mol } L^{-1})$$

$$K_{sp} = [Ag^+]\,[Cl^-]$$

$$K_{sp} = S^2$$

In NaCl solution

$$NaCl \longrightarrow Na^+ + Cl^-$$

$$C \qquad C \qquad C$$

Let solubility of AgCl in the presence of NaCl solution is S' mol L^{-1}

$$AgCl \rightleftharpoons Ag^+ + Cl^-$$

$$S' \qquad S'+C$$

$$K_{sp} = [Ag^+]'\,[Cl^-]'$$

$$K_{sp} = S'\,(S' + C) = S'^2 + S'C \qquad \text{(Neglecting the higher power terms of S')}$$

$$K_{sp} = S'\,C$$

$$\boxed{S' = \frac{K_{sp}}{C}}$$

10.5 SIMULTANEOUS SOLUBILITY :

When two sparingly soluble salts are added in water simultaneously, there will be two simultaneous equilibria in the solution.

10.6 SOLUBILITY IN APPROPRIATE BUFFER SOLUTIONS :

Appropriate buffer means that the components of buffer should not interfere with the salt or only H^+ or OH^- ions should be interacting with the ions of the salt.

10.7 EFFECT ON SOLUBILITY BECAUSE OF COMPLEX FORMATION :

Solubility of AgCl in aqueous NH_3 is roughly 10,000 times as its solubility in water, due to complex formation.

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

$$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^{\,+}(aq)\ ; \qquad K_{eq} = K_{stability} = K_{formation}$$

and $$\frac{1}{K_{stability}} = K_{dissociation} = K_{instability}$$

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10.8 CONDITION OF PRECIPITATION /IONIC PRODUCT (IP OR Q_{SP}) :

- Ionic product (IP) of an electrolyte is defined in the same way as K_{sp}. The only difference is that ionic product expression contains the initial concentration of ions or the concentration at any time whereas the expression of K_{sp} contains only equilibrium concentration. Thus, for AgCl.

  $IP = [Ag^+]_i\,[Cl^-]_i$ and $K_{sp} = [Ag^+]_{eq}.[Cl^-]_{eq}$

- Ionic product changes with concentration but K_{sp} does not.
- To decide whether an ionic compound will precipitate, its K_{sp} is compared with the value of ionic product. The following three cases arise :

  (i) $IP < K_{sp}$: The solution is unsaturated and precipitation will not occur.

  (ii) $IP = K_{sp}$: The solution is saturated and solubility equilibrium exists.

  (iii) $IP > K_{sp}$: The solution is supersaturated and hence precipitation of the compound will occur.

  Thus, a salt is precipitated when its ionic product exceeds the solubility product of the salt.

10.9 SELECTIVE PRECIPITATION :

When the k_{sp} values differ then one of the salt can be selectively precipitated.

Ex. 39. (i) $Al_2(SO_4)_3(s) \rightleftharpoons 2Al^{+3}(aq) + 3SO_4^{-2}(aq)$

$K_{sp} = 2^2 \times 3^3 \times (S)^{2+3} = 4 \times 27 \times S^5 = 108\,S^5$

(ii) $Na_2KPO_4(s) \rightleftharpoons 2Na^+(aq) + K^+(aq) + PO_4^{-3}(aq)$

$K_{sp} = 2^2 \times 1^1 \times 1^1 (S)^{2+1+1} = 4S^4$

(iii) $NaKRbPO_4(s) \rightleftharpoons Na^+(aq) + K^+(aq) + Rb^+(aq) + PO_4^{-3}(aq)$

$K_{sp} = 1^1 \times 1^1 \times 1^1 \times 1^1 \times (S)^{1+1+1+1} = S^4$

Ex. 40. If solubility product of the base $M(OH)_3$ is 2.7×10^{-11}, the concentration of OH^- will be

(1) 3×10^{-3} ***(2) 3×10^{-4}*** ***(3) 10^{-3}*** ***(4) 10^{-11}***

Answer :(3)

Ex. 41. The solubility of $BaSO_4$ in water is 1.07×10^{-5} mol dm^{-3}. Estimate its solubility product.

Sol. *Solubility equilibrium for $BaSO_4$ is*

$$BaSO_{4(s)} \rightleftharpoons Ba^{2+}_{(aq)} + SO^{2-}_{4(aq)}$$

Now, $S = 1.07 \times 10^{-5}\ M$

Hence, $K_{sp} = (1.07 \times 10^{-5})^2 = 1.145 \times 10^{-10}$

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Ex. 42.The solubility product of AgBr is 5.2×10^{-13}. Calculate its solubility in mol dm^{-3} and g dm^{-3}. (Molar mass of AgBr. = 187.8 g mol^{-1})

Sol. *The solubility equilibrium of AgBr is*

$$AgBr_{(s)} \rightleftharpoons Ag^{+}_{(aq)} + Br^{-}_{(aq)}$$

The molar solubility S of AgBr is given by

$$S = \sqrt{K_{sp}} = \sqrt{5.2 \times 10^{-13}} = 7.2 \times 10^{-7} \text{ mol } dm^{-3}$$

The solubility in g dm^{-3} = molar solubility (mol dm^{-3}) × molar mass (g mol^{-1})

$$= 7.2 \times 10^{-7} \times 187.8 = 1.35 \times 10^{-4} \text{ g } dm^{-3}$$

Ex. 43.What is the maximum volume of water required to dissolve 1 g of calcium sulphate at 25°C. For calcium sulphate, $K_{sp} = 9.0 \times 10^{-6}$.

Sol. $CaSO_4(aq) \rightleftharpoons Ca^{2+}(aq) + SO_4^{2-}(aq)$

If S is the solubility of $CaSO_4$ in moles L^{-1}

$K_{sp} = [Ca^{2+}] \times [SO_4^{2-}] = S^2$

$$\therefore S = \sqrt{K_{sp}} = \sqrt{9.0 \times 10^{-6}}$$

$= 3 \times 10^{-3}$ mol L^{-1}

$= 3 \times 10^{-3} \times 136$ g $L^{-1} = 0.408$ gL^{-1}

For dissolving 0.408 g of $CaSO_4$ water required = 1 L

$$\therefore \quad \text{For dissolving 1g } CaSO_4 \text{ water required} = \frac{1}{0.408} \text{L} = 2.45 \text{ L}$$

Ex. 44.Equal volumes of 0.04 M $CaCl_2$ and 0.0008 M Na_2SO_4 are mixed. Will a precipitate form? K_{sp} for $CaSO_4 = 2.4 \times 10^{-5}$

Sol.

	$CaCl_2$	+ Na_2SO_4	→ $CaSO_4$	+ $2NaCl$
Millimole added	$0.04\,V$	$0.0008 \times V$	0	0

Suppose V mL of both are mixed

$$\therefore \quad [Ca^{2+}] = \frac{0.04\,V}{2V}$$

$$[SO_4^{2-}] = \frac{0.0008\,V}{2V}$$

$$\therefore \quad [Ca^{2+}][SO_4^{2-}] = \frac{0.04\,V}{2V} \times \frac{0.0008\,V}{2V} = 8 \times 10^{-6}$$

Thus, $[Ca^{2+}][SO_4^{2-}]$ in solution < K_{sp}

$8 \times 10^{-6} < 2.4 \times 10^{-5}$

$\therefore$ *$CaSO_4$ will not precipitate.*

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Ex. 45. Calculate simultaneous solubility of silverthiocyanate and sliver bromide in water given that k_{sp} of silver thiocyanate = 10^{-12} and k_{sp} of silver bromide = 5×10^{-13} respectively.

Sol. *Let the solubility of AgSCN be x and that of AgBr is y, then*

$$AgSCN \rightleftharpoons Ag^+ + SCN^- \qquad AgBr \rightleftharpoons Ag^+ + Br^-$$

$$x + y \quad x \qquad x + y \quad y$$

$10^{-12} = x(x + y)$ ..... (i)

$5 \times 10^{-13} = y(x + y)$ ..... (ii)

On solving we get, $x = 2y$

So $y = 4.08 \times 10^{-7}$ *and* $x = 8.16 \times 10^{-7}$

Ex. 46. What $[H^+]$ must be maintained in saturated H_2S(0.1 M) to precipitate CdS but not ZnS, if $[Cd^{2+}] = [Zn^{2+}] = 0.1$ initially ?

$K_{sp} = (CdS) = 8 \times 10^{-27}$

$K_{sp} = (ZnS) = 1 \times 10^{-21}$

$K_a = (H_2S) = 1.1 \times 10^{-21}$

Sol. *In order to prevent precipitation of ZnS*

$[Zn^{2+}][S^{2-}] < K_{sp}(ZnS) = 1 \times 10^{-21}$

(ionic product)

or $(0.1)[S^{2-}] < 1 \times 10^{-21}$

or $[S^{2-}] < 1 \times 10^{-20}$

This is the maximum value of $[S^{2-}]$ before ZnS will precipitate. Let $[H^+]$ to maintain this $[S^{2-}]$ be x.

Thus for $H_2S \rightleftharpoons 2H^+ + S^{2-}$

$$K_a = \frac{[H^+]^2[S^{2-}]}{[H_2S]} = \frac{x^2(1 \times 10^{-20})}{0.1} = 1.1 \times 10^{-21}$$

or $x = [H^+] = 0.1\ M$

$\therefore$ *No ZnS will precipitate at a concentration of H^+ greater than 0.1 M*

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Ex. 47. What must be the concentration of aq. NH_3(eq.) which must be added to a solution containing 4×10^{-3} M Ag^+ and 0.001 M NaCl, to prevent the precipitation of AgCl.

Given that $K_{sp}(AgCl) = 1.8 \times 10^{-10}$ and the formation constant of $[Ag(NH_3)_2]^+$ is $K_{formation} = \frac{10^8}{6}$.

Sol. *Calculate silver ion concentration which can be allowed to remain in the solution,*

$1.8 \times 10^{-10} = [Ag^+][Cl^-]$

$[Ag^+] = \frac{1.8 \times 10^{-10}}{0.001} = 1.8 \times 10^{-7}$ M,

This quantity is so small that almost all the Ag^+ ion will be consumed.

Ag^+	+	$2NH_3$	$\rightleftharpoons$	$[Ag(NH_3)_2]^+$	$K = \frac{10^8}{6}$
4×10^{-3}		b		0	
1.8×10^{-7}		$(b - 8 \times 10^{-3})$		4×10^{-3}	$K = \frac{10^8}{6} = \frac{4 \times 10^{-3}}{1.8 \times 10^{-7} \times (b - 8 \times 10^{-3})^2}$

$\Rightarrow \quad b = 0.0445$

Ex. 48. 0.10 mol sample of $AgNO_3$ is dissolved in one litre of 2.00 M NH_3. Is it possible to form AgCl(s) in the solution by adding 0.010 mol of NaCl ?

$(K_{sp(AgCl)} = 1.8 \times 10^{-10}, K_{f[Ag(NH_3)_2^+]} = 1.6 \times 10^7)$

Sol.

Ag^+	+	$2NH_3$	$\rightleftharpoons$	$[Ag(NH_3)_2^+]$
0.10 M		2.00		0
0.10–0.10		(2–0.20 M)		0.10 M
= 0		= 1.80 M		

It is assumed that all Ag^+ ions have been complexed and only x amount is left

$K_f = \frac{[Ag(NH_3)_2^+]}{[Ag^+][NH_3]^2} \Rightarrow 1.6 \times 10^7 = \frac{0.10}{x(1.80)^2}$

$\therefore \quad x = 1.93 \times 10^{-9}$ M $= [Ag^+]$ *undisolved*

$[Cl^-] = 1.0 \times 10^{-2}$ M

$\therefore \quad [Ag^+][Cl^-] = 1.93 \times 10^{-9} \times 1.0 \times 10^{-2} = 1.93 \times 10^{-11} < 1.8 \times 10^{-10} [K_{sp(AgCl)}]$

Hence, AgCl (s) will not precipitate.

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Ex. 49.What is the concentration of Ag^+ ions in 0.01 M $AgNO_3$ that is also 1.0 M NH_3 ? Will AgCl precipitate from a solution that is 0.01 M $AgNO_3$, 0.01 M NaCl and 1 M NH_3?

$K_d(Ag[NH_3]_2^+) = 5.88 \times 10^{-8}$; $K_{sp}(AgCl) = 1.8 \times 10^{-10}$.

Sol. *Let us first assume that 0.01 M $AgNO_3$ shall combine with 0.02 NH_3 to form 0.01 M $Ag(NH_3)_2^+$ and the consider its dissociation.*

$AgNO_3$	+ $2NH_3$	⟶ $Ag(NH_3)_2^+$	...Initial conc.
0.01 M	1 M	0	
0	(1–0.02)=0.98M	0.01 M	...at eq. conc.

$$Ag(NH_3)_2^+ \rightleftharpoons Ag^+ + 2NH_3$$

$(0.01 - x)$	x	$(0.98 + 2x)$
$= 0.01$ M		≈ 0.98 M
		....Equib. conc.

Since $x <<< 1$

$$K_d = \frac{[Ag^+][NH_3]^2}{[Ag(NH_3)_2^+]} = 5.88 \times 10^{-8}$$

$$\therefore \quad [Ag^+] = \frac{5.88 \times 10^{-8} \times 0.01}{(0.98)^2} = 6.12 \times 10^{-10}\ M$$

Further, ionic product of AgCl $= [Ag^+]\,[Cl^-] = (6.12 \times 10^{-10})\,(0.01) = 6.12 \times 10^{-12}$

Because the ionic product is smaller than $K_{sp} = 1.8 \times 10^{-10}$*, no precipitate should form.*

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